

2022

From
Codersera

Turtle
import turtle

wn = turtle.Screen()
alex = turtle.Turtle()

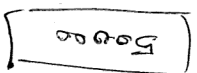
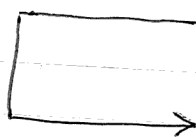
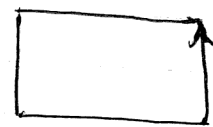
alex.forward(150)

alex.left(90)

alex.forward(75)

- # creates a graph window
- # create a turtle called alex
- # tell alex to move forward by 150 units
- # turn by 90 degrees
- # complete the second side of rectangle.

alex.salary = 50000
print(alex.salary)
alex.left(90)
alex.forward(150)
alex.forward(75)



80000

8

9

10

11

12

1

2

3

4

5

6

7

8

الأحد 31
يوم - يوليو

2022 Week 31, Month 7 Day: 212 / 153 2 Muharram 1 / 1444

Sunday 31
Dimanche
July - Juillet

zip Function

zip(iter1, iter2 [...])
→ zip object

list1 = [1, 2, 3]
list2 = [10, 20, 30]
print(zip(list1, list2)) ⇒ <zip object at 0x...>
print(list(zip(list1, list2))) ⇒ [(1, 10), (2, 20), (3, 30)]
print(list(zip('abc', list1, list2))) ⇒ [(a, 1, 10), (b, 2, 20), (c, 3, 30)]

Stacks

Last in - Last out → LIFO
browsing-session = []
browsing-session.append(1)
browsing-session.append(2)
browsing-session.append(3)
print(browsing-session) ⇒ [1, 2, 3]

7 JUL

Saturday 30

Samedi

July - Juillet

1 Muharram 1 / 1444

Day: 211 / 154

2022

Week 31, Month 7

1 محرم 1444 / 1

تقویم - يوليو



Filter Function

Filter(Function or none, iterable)
→ Filter object

items = [

("product1", 10),
("product2", 9),
("product3", 12),

] x = Filter(lambda item: item[1] >= 10, items)
Print(x) → Filter object at 0x...>

Filtered = list(Filter(lambda item: item[1] >= 10, items))
Print(Filtered) → [('product1', 10),
('product2', 12)]

[Expression for item in items]

prices = list(map(lambda item: item[1], items))
prices = [item[1] for item in items]

Filtered = list(Filter(lambda item: item[1] >= 10, items))
Filtered = [item for item in items if item[1] >= 10]

Map Function

map(func, *iterables)

```

items = [
    ("product1", 10),
    ("product2", 9),
    ("product3", 12),
]

prices = []
for item in items:
    prices.append(item[1])

print(prices) => [10, 9, 12]
    
```

```

items = [
    ("product1", 10),
    ("product2", 9),
    ("product3", 12),
]

x = map(lambda item: item[1], items)
print(x) => <map object at 0x...>
    
```

```

x = map(lambda item: item[1], items)
for item in x:
    print(item) => 10
    9
    12

prices = list(map(lambda item: item[1], items))
print(prices) => [10, 9, 12]
    
```



```

!Items = [
    ("product1", 10),
    ("product2", 9),
    ("product3", 12),
]

def sort_item(item):
    return item[1]

[print(item)
 for item in sorted(Items, key=sort_item)]

```

Lambdas

$\text{items} = [$

- "(product1, 10),"
- "(product2, 9),"
- "(product3, 12),"

$] \quad]$

$\text{item, sort(key = lambda item: item[1])}$

$\Rightarrow [(product2, 9),$

$(product1, 10),$

$(product3, 12)]$

Print(items)

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البريد
البريد
البريد
البريد

items.sort() $\Rightarrow$ [('Product', 10),
('Product', 9),
('Product', 12)]

items = [
('Product', 10),
('Product', 9),
('Product', 12)]

numbers = [3, 51, 2, 8, 6]
print(sorted(numbers)) $\Rightarrow$ [2, 3, 6, 8, 51]
print(numbers) $\Rightarrow$ [3, 51, 2, 8, 6]
print(sorted(numbers, reverse=True)) $\Rightarrow$ [51, 8, 6, 3, 2]
print(numbers) $\Rightarrow$ [3, 51, 2, 8, 6]

numbers = [3, 51, 2, 8, 6]
numbers.sort()
print(numbers) $\Rightarrow$ [2, 3, 6, 8, 51]
numbers.sort(reverse=True)
print(numbers) $\Rightarrow$ [51, 8, 6, 3, 2]

Sorting List

Tuesday
Mardi!

July - Juliet

27 DhuHijjah 12 / 1443

Day: 207 / 158

2022

Week30, Month7

١٦٣٣١/٢١ ٢١٢٢

$$= \frac{1}{2} \left(\frac{1}{2} + \frac{1}{2} \right) = \frac{1}{2}$$

٢٢

Print(letters) => 0
1
2

Print(letters[0], letters[1]) => 0 a
1 b
2 c

letters = ['a', 'b', 'c']
For index, letter in enumerate(letters):
Print(index, letter) => 0 a
1 b
2 c

Adding / Removing items

letters = ['a', 'b', 'c']

#Add

letters.append('d')

Print(letters) => ['a', 'b', 'c', 'd']

letters.insert(0, 'u')

Print(letters) => ['u', 'a', 'b', 'c', 'd']

(X) first, second = numbers ⇒
 (too many values to unpack)

numbers = [1, 2, 3, 4, 4, 4, 4, 4]
 first, second, other = numbers
 print(first) ⇒ 1
 print(other) ⇒ [3, 4, 4, 4, 4, 4, 4]

numbers = [1, 2, 3, 4, 4, 4, 4, 9]
 first, other, last = numbers
 print(first, last) ⇒ 1 9
 print(other) ⇒ [2, 3, 4, 4, 4, 4]

Looping over Lists

letters = ['a', 'b', 'c']
 for letter in letters:
 print(letter) ⇒ a
 b
 c

letters = ['a', 'b', 'c']
 for letter in enumerate(letters):
 print(letter) ⇒ (0, 'a')
 (1, 'b')
 (2, 'c')

Saturday
Samedi!

23

2022

Accessing Items

$$[n p_n', n c_n', n q_n', n b_n] = 52401$$

```
print(letters[0])
```

```
print('letters[-1]') => d
```

$$\text{letters}[0] = 'A'$$

```
print(letters) => ['A', 'B', 'C', 'D']
```

```
print(letters[0:3]) ➡ ['a', 'b', 'c']
```

```
print(letters[3]) => ['A', 'b', 'c']
```

```
print(letters[0:]) ⇒ ['A', 'b', 'c', 'd']
```

```
print('letters: ') => ['A', 'b', 'c', 'd']
```

```
print(letters[1:2]) => ['A', 'c']
```

numbers = list(range(20))

print(numbers) $\Rightarrow$ [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19]

$[12, 13, 14, 15, 16, 17, 18, 19]$

Print (numbers [12] $\Rightarrow$ [0, 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 32, 34, 36, 38, 40, 42, 44, 46, 48, 50])

```
print(numbers[::-1]) # [19, 18, 17, 16, 15, 14, 13, 12, 11, 10, 9, 8, 7, 6, 5, 4, 3, 2, 1, 0]
```

Unpacking Lists

numbers = [1, 2, 3]

First number [0]

Sec number [17]

third = number[2]

First, second, third = numbers

$$\text{letters} = [a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z]$$

matrix $z = [[0, 1], [2, 3]]$

$$Z_{eros} = [0] \times 100$$

Print(zeros)

0 1 0 1 0 1 0 1 $\leftarrow$

[010101]

letters = [a, b, c]

Matrix = $[[0, 1], [2, 3]]$

$$Zeros = [0] * 5$$

Combined = Zeros + letters

$\text{print}(\text{combined}) \Rightarrow [0, 0, 0, 0, 0, 0, 0, 0, 0]$

$$\text{numbers} = \text{listRange}(201)$$

Print (numbers)

$$\Rightarrow [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19]$$

Chars = 151 (middle world)

private(chars) ← [H, e, l, l, o, w, o, r, l, d]

$\text{PrintLen}(\text{chars}) \geq n$

July - Juillet

Thursday 21 Juedi

22 DhuHijjah 12 / 1443

Day: 202 / 163

2022

Week 29, Month 7

٢٢ ذو الحجة ١٤٤٣ / ١٢

تموز - يوليو

٢١ جمادى الأولى

Print(Fizz-Buzz(15)) => FizzBuzz
Print(Fizz-Buzz(7)) => 7

return input

return "Fizz"

if input % 5 == 0:

return "Fizz"

if input % 3 == 0:

return "FizzBuzz"

if (input % 3 == 0) and (input % 5 == 0):

def Fizz-Buzz(input):

مع فليزر:

لنفسه

في فليزر and فليزر

Fizz-Buzz (15) => Fizz

Print(Fizz-Buzz(5)) => Buzz
Print(Fizz-Buzz(15)) => Fizz

return input

return "FizzBuzz"

if (input % 3 == 0) and (input % 5 == 0):

return "Fizz"

if input % 5 == 0:

return "Fizz"

if input % 3 == 0:

def Fizz-Buzz(input):

$F_n + F_{n-1}$
 F_5
 F_{10}
 (Depag) F_{10}
 F_{11}
 $shift + F_5$
 $shift + F_{11}$

```

def FizzBuzz(input):
    if input % 3 == 0:
        result = "Fizz"
    else:
        result = "Buzz"
    return result

```

Print(FizzBuzz(5)) ⇒ Buzz

```

def FizzBuzz(input):
    if input % 3 == 0:
        return "Fizz"
    else:
        return "Buzz"

```

Print(FizzBuzz(5)) ⇒ Buzz

def greet():
message = "a"

print(message) ⇒ a

message = "a"

def greet(name):
message = "a"

greet("a")
print(message) ⇒ a

message = "a"

def greet(name):
global message
message = "a"

greet("a")
print(message) ⇒ a

def mult(p1, p2):
return p1 * p2

total = 1
for number in numbers:
total *= number

print(mult(1, 2))
print(mult(1, 2))
print(mult(1, 2))

Monday
Lundi

18

July - Juillet

19 DhuHijjah 12 / 1443

Day: 199 / 166

2022

Week 29, Month 7

19 ذو الحجة 1443 / 12

تموز - يوليو

الأثنين ١٨

def multiply(*numbers)

for number in numbers:

print(number)

multiply(2,3,4,5) => 2
3
4
5

def multiply(*numbers):

total = 1

for number in numbers:
total *= number

return total

print(multiply(2,3,4,5)) => 120

def save_user(*users):

print(user)

save_user(id=1, name="John", age=22)

=> {id:1, name:"John", age:22}

def save_user(*user):

print(user["name"])

save_user(id=1, name="John", age=22)

=> [John]

LL

Dimanche

July - Juliet

18 DhulHijjah 12 / 1443

Day: 198 / 167

Week29, Month7

2022

١٦٣٩ / ١٢ سنة الحجة ١٢٨٠

$$- \frac{1}{2} \frac{d}{dt} \left(\frac{1}{2} \frac{d}{dt} \right) = - \frac{1}{2} \frac{d}{dt} \left(\frac{1}{2} \frac{d}{dt} \right)$$

$\text{Greet}(u, \text{Mosh}) \Rightarrow \text{H! Mosh}$
 $\text{Print}(\text{Greet}(u, \text{Mosh})) \Rightarrow \text{H! Mosh}$ } None

Def: increment(number by)
return number + by

$\{ \text{result} = \text{increment}(2, 1) \} \Rightarrow 3$
 $\{ \text{print}(\text{result}) \} \Rightarrow 3$
 $\text{print}(\text{increment}(2, 1)) \Rightarrow 3$
 $\text{print}(\text{increment}(2, \text{bg} = 1)) \Rightarrow 3$

```
print (increment(2))  $\Rightarrow$  3
print (increment(2,5))  $\Rightarrow$  7
```

```
def increment(number, by=1):
    return number + by
```

Def mult: $\text{ply}(X, Y) = \text{return } X * Y$

mult: p1y (2,3) Xargs

def mult(pid, numbers):
 print(numbers)

Print (headers)

$$\text{mult. pty} (2, 3, 4, 5) \Rightarrow (2, 3, 4, 5)$$

9 L

July - June

2202

Day: 197 / 168

21/2331 21/2331

$$\frac{1}{2} = \frac{1}{2}$$

```
def greet(first_name, last_name):
```

```
print(f"Hi! {first-name} {last-name}")
print("Welcome aboard")
```

$\equiv \text{gred}(\text{Mosh}, \text{Hamdan})$

It! Mesh Hame welcome abroad

$$\Leftarrow ({}^n\text{Smith}, {}^n\text{John}) \text{ greet}({}^n\text{John})$$

Hi John Smith
Welcome aboard

It is $\in \mathbb{N}$ valuable functions in *

123

Def- greet (name):
print(f"Hi! {name}")

```
greet(name);
print(f"it's {name}'s")
```

```
def get_greeting(name):
    return f'Hi! {name}'
```

message = get-greeting("Mosh")

Print (message)

$$[File = open('x.txt', 'a')] \quad \text{File = open('x.txt', 'a')}$$

6. File.write(message)

ॐ नमो भगवते वासुदेवाय - ॐ नमो भगवते वासुदेवाय



```
print(type(range(5))) > <class 'int'>
print(type(range(5))) > <class 'range'>
```

while True

Command = input('n > v')

```
Print("Echo", command)
```

! If command $\text{lower}(z) = \text{quf}$;

break

Count = 0

For number in range(1, 10):
if number%2==0:

$$\text{Count} = \text{Count} + 1$$

Print(number)

Print ("we have {count} even numbers")

2 4 6 8

we have 4 even numbers

Functions

def- greet():

Print (n!) there

print- ("welcome abroad")

Greif (1)

It: three

welcome abroad

Thursday 14

Jeddi

July - Juillet

15 DhuHijjah 12 / 1443

Day: 195 / 170

2022

Week 28, Month 7

10 نون الحجة 1443 / 12

تموز - يوليو

13 جمادى

```

    successful = false
    for number in range(3):
        print("Attempt")
        if successful:
            print("Successful")
            break
        else:
            print("Attempt 3 times and failed")
    print("Attempt 3 times and failed")
    print("Successful string is valid")
    print("Successful")

```

Attempt 3 times and failed

Attempt
Attempt
Attempt

جملتين

for x in range(5):
 for y in range(3):
 print(f"x {x} y {y}")

(0,0)
(0,1)
(0,2)
(1,0)
(1,1)

Wednesday 13

Mercredi

July - Juillet

14 DhulHijjah 12/1443

Day: 194 / 171

2022

Week 28, Month 7

15 ذو الحجة 1443/12

تموز - يوليو

الأول

For number in range (1,4):
Print("Attempt", number + (number * n.4))

⇒ Attempt 1.
Attempt 2..
Attempt 3...

For number in range (1,10,2):
Print("Attempt", number + number * n.4)

⇒ Attempt 1.
Attempt 3...

5.....

7.....

9.....

successful = True

For number in range (3):
Print("Attempt")

! If successful:

Print("successful")

break

⇒ Attempt
successful

تموز - يوليو

Tuesday 12

Mardi

July - Juillet

13 Dhu'l-Hijjah 12 / 1443

Day: 193 / 172

2022

Week 28, Month 7

13 ذو الحجة 12 / 1443

تموز - يوليو

12

if- lo = "lo" :

Print("a")

elif "bag" > "apple" and "bag" < "cat" :

Print("b")

else:

Print("c")

For number in range(3):

Attempt
Attempt
Attempt

For number in range(3):

Print("Attempt", number) => Attempt 0
Attempt 1
Attempt 2

Print("Attempt", number+1) => Attempt 1
Attempt 2
Attempt 3

Print("Attempt", number+1 + (number+1)*"n") => Attempt 1,
Attempt 2 ..
Attempt 3 ...

student = True
 if not student:
 print("Eligible")
 else:
 print("Not eligible")
 } ⇒ Not eligible

False
 } student
 Eligible
 if (high-income and good-credit) and not student

if (high-income and good-credit) and not student
 print("Eligible")
 else:
 print("Not eligible")

65, 18 in age
 age = 22
 if age >= 18 and age < 65:
 print("Eligible")

if 18 <= age < 65:
 print("Eligible")

Sunday 10

Dimanche

July - Juillet

11 DhulHijjah 12/1443

Day: 191 / 174

2022

Week 28, Month 7

11 ذو الحجة ١٢ / ١٤٤٣

تموز - يوليو

الأحد ١٠

age = 22
if age >= 18:
message = "Eligible"
else:
message = "Not eligible"
print(message)
age = 22
message = "Eligible" if age >= 18 else "Not eligible"
print(message)
Not eligible
Eligible
12 age = 12
11 age = 22
10 message = "Eligible" if age >= 18 else "Not eligible"
9 message = "Eligible"
8 if age >= 18:
message = "Eligible"
else:
message = "Not eligible"
print(message)

Not eligible
Eligible
if high-income and good-credit = True
good-credit = True
if high-income and good-credit:
print("Eligible")
high-income = False
good-credit = True
if high-income and good-credit:
print("Eligible")
else:
print("Not eligible")
or and
Eligible

high-income = False
good-credit = True
if high-income and good-credit:
print("Eligible")
else:
print("Not eligible")
or and
Eligible

July - Juillet

Saturday
Samedi

9

تموز - يوليو

9

Week 28, Month 7

2022

Day: 190 / 175

10 ذو الحجة 1443 / 17

10 DhuHijjah 12 / 1443

```
temperature >= 35
if temperature > 30 :
    print("It's warm")
    print("Drink water")
    print("Done")
```

It's warm
Drink water
Done

temperature >= 15
Done

```
temperature = 15
if temperature > 30 :
    print("It's warm")
    print("Drink water")
elif temperature > 20 :
    print("It's nice")
else:
    print("It's cold")
    print("Done")
```

It's cold
Done

الطريق إلى
while command lower 12 uqil 2

for loop while condition
if command lower 12 uqil 2

break

Comparison operators

10 > 3 ⇒ True

10 >= 3 ⇒ True

10 < 20 ⇒ True

10 == 10 ⇒ True

10 >= 10 ⇒ False

10 / 2 > 10 ⇒ True

"bag" > "apple" ⇒ True

"bag" == "BAG" ⇒ False

ord("b") > 98

ord("B") > 66

bool("False") == True

for x in "python":
print(x)

for x in [1, 2, 3, 4]:
print(x)

number = 100
while number > 0:
print(number)
number //= 2

while command != "quit":
command = input(" ")
print("ECHO", command)

2+2 ==> ECHO 2+2
3*2 ==> ECHO 3*2
QUIT ==> ECHO QUIT

quit ==> QUIT ==> ECHO QUIT
while command != "quit" and command != "QUIT":

Type conversion

$x = \text{input}("x: ")$

$y = x + 1$

$\Rightarrow \text{Error}$

int x و str y
 هنا المشكلة

int(x)
 float(x)
 bool(x)
 str(x)

$\text{print}(\text{type}(x)) \Rightarrow < \text{class 'str'} >$

$x = \text{input}("x: ")$
 $y = \text{int}(x) + 1$

$\text{print}(f"x: {x}, y: {y}")$

$\Rightarrow x: 1 \Rightarrow x: 1, y: 2$

False or True

$\hookrightarrow \text{Python3 and 2}$

$\text{bool}(0) \Rightarrow \text{False}$

$\text{bool}(1) \Rightarrow \text{True}$

$\text{bool}(-1) \Rightarrow \text{True}$

$\text{bool}(5) \Rightarrow \text{True}$

$\text{bool}(\text{num}) \Rightarrow \text{False}$

* google → (Python 3 math module)
 → e.g. docs.python.org → Number theoretic and — Function

Working with numbers
 import math
 print(round(2.9)) → 3
 print(abs(-2.9)) → 2.9
 print(math.ceil(2.2)) → 3

$x = 10$
 $x = x + 3$
 $\Rightarrow x = 13$

Numbers
 $x = 1$
 $x = 1.1$
 $x = a + bi$
 $x = 1 + 2j$
 print((1+3)) → 13
 print(10-31) → 7
 print(10*3) → 30
 print(10/3) → 3.33333335
 print(10/3) → 3
 print(10%3) → 1
 print(10*3) → 30

Monday

Lundi

July - Juillet

4

5 DhuHijjah 12 / 1443

Day: 185 / 180

2022

Week 27, Month 7

1443 / 12 / 12

تموز - يوليو

الأربعاء

3

full = f" {len(first)} {last}"
print(full) => 4 Heurdaani
full = f" {len(first)} {last}"
print(full) => 4 4

String Methods
course = "python programming"
(print(course)) => python programming
print(course.upper()) => PYTHON PROGRAMMING
print(course.lower()) => python programming
print(course.strip()) => python programming
print(course.strip()) => python programming
course-capital = course.upper()

print(course.istrip())
print(course.rstrip())
print(course.find("pre")) => 9
print(course.find("pre")) => -1
print(course.replace("pre", "py")) => python programming
print("pre" in course) => True
print("swift" not in course) => True

Sunday
Dimanche
3

July - Juillet

4 DhulHijjah 12 / 1443

Day: 184 / 181

2022

Week 27, Month 7

3 ذو الحجة 1443 / 12

تموز - يوليو

الأحد

7 JUL

```
course = "Python programming"
print(len(course)) ➔ 18
print(course[0]) ➔ P
print(course[-1]) ➔ g
print(course[0:3]) ➔ Pyt
print(course[0:]) ➔ Python programming
print(course[3:]) ➔ pyt
print(course[:7]) ➔ python programming
```

```
course = 'python "programming"'
print(course) ➔ python "programming"
course = "python 'programming'"
print(course) ➔ python 'programming'
course = 'python \n programming'
print(course) ➔ python \n programming
course = "Python /n programming"
print(course) ➔ Python /n programming
```

first = "Mosh"

last = "Hamedani"

full = first + " " + last

print(full) ➔ Mosh Hamedani

full = f"{first} {last}"

print(full) ➔ Mosh Hamedani

Saturday
Samedi

2

July - Juillet

3 DhuHijjah 12 / 1443

Day: 183 / 182

2022

Week 27, Month 7

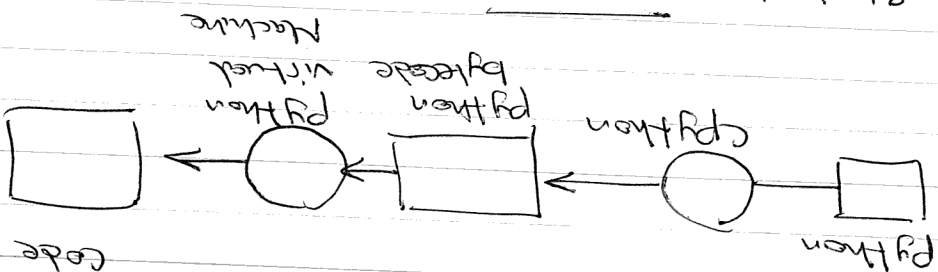
1443 / 12 / 12

تونس - تونس

السنة

٢

Machine
code



Student-Count = 1000
Print(Student-Count) ⇒ 1000

Student-Count = 1000
Rating = 4.99
is-Published = False
Course-Name = "Python Programming" String
Message = null

Hi John!

This is Mosh from [codecademy.com](https://www.youtube.com/channel/UCWv91-G3_vk0pQm3vZ0L8Yw)

blab blab blab

(greeting course & level: 1000)

Friday
Vendredi
July - Juillet

2 DhulHijjah 12 / 1443

Day: 182 / 183

2022 Week 26, Month 7

1443 / 12 / 12

تموز - يوليو
الجمعة

VS Code

Mosh. Copying

Tools -> Editor -> VScode - Atom - Sublime
IDE -> Pycharm

*code.visualstudio.com
[] Search
[] Enable linking

* Google -> PEP8 -> pep8
[] VScode -> [] Code runner
[] VScode -> [] Install

* ctrl + alt + n

* view -> open view
Format Document -> install (autopep8)
Setting -> Preferences -> settings -> Format on save

* user settings -> ... -> open

* Code-runner: executorsnap
in file, settings -> user settings -> autopep8
"python": "python3"

Implementations
Python -> Java
IronPython -> C#
PyPy -> subset of Python

Python

Friday
Vendredi
20

May - Mai

19 Shawal 10 / 1443

Day: 140 / 225

2022

Week 20, Months

19 شوال 1443/10

أيار - مايو

٢٠
الجمعة

8

NSString = NSString

m = 3.1416

9

مستطيل في دائرة

① apocv

⑤ kaken

⑦ PyHesseraat

11

مستطيل في دائرة
مستطيل في دائرة
مستطيل في دائرة

12

مستطيل في دائرة

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18 Shawal 10/1443 Day: 139 / 226 2022 Week20, Month5 18 شوال 1443/10

May - Mai

Thursday 19 Jendi

19 جمادى الثاني
ايار - مايو

٨ يونيو

٩. Lists -
الصفوف (المصفوفات) -
lambda (دالة لAMBDA) -
التuple (المتuple) -
Dictionary set, tuple, list -
في الصفوف المجمعة -

١١. C-Like = C-
C, C++, Java, C# هي

١٢. Packages -
Classes -
الصفوف

١. الصفوف (المصفوفات) -
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١٠٠. الصفوف (المصفوفات) -

٣. Read a/b
Sa = input("a = "); a = int(Sa)
Sb = input("b = "); b = int(Sb)

٥. # compare GCD of a and b
while a != b:
if a > b:

a = a - b
else:

b = b - a

8. g = a
print("GCD = " + str(g))

```
sc = input("input Boolean? "); c = bool(sc)
```

3# Peak Boolean

$$sb = \text{input}(\text{"input Float"}), b = \text{Float}(sb)$$

2nd Year Float

$8a = \text{input}(\text{"input integer"})$; $a = \text{int}(8a)$

read Integer

$\text{args} \leftarrow (\text{bool}, \text{float}, \text{int})$ و ints في input()

10 n n n

1661

Python is an interpreted, high level and general-purpose programming language. created by Guido van Rossum and first released in 1990.

(Handwritten signature)

نك

```

a=5, b=7, c=8, g=1,
print("a,b,c", a,b,c),
print("gcd=" + str(g)).
    
```

$a=b=c=d=8$
 # تعريف المتغيرات

{ +, -, *, /, %, ^ } : العمليات
 * التعريفات والعمليات

$x \text{ op } y$
 ← وهو ما يعني $x \text{ op } y$

نك

$a = -8$
 $a, b = 56$
 $a + b$
 $b * = 3$

Expressions
 ←
 ()
 *
 /
 %
 +

① تعريف المتغيرات
 ← وهو ما يعني $x \text{ op } y$

بما اننا نعلم ان float هو نوع من float ...

في هذه الحالة...

نلاحظ اننا نعلم ان float هو نوع من float ...

نلاحظ اننا نعلم ان float هو نوع من float ...

P ()

E **

M *

D /

→

%

A +

→

S -

if ...

if-else ...

if expression:
instructionBlock

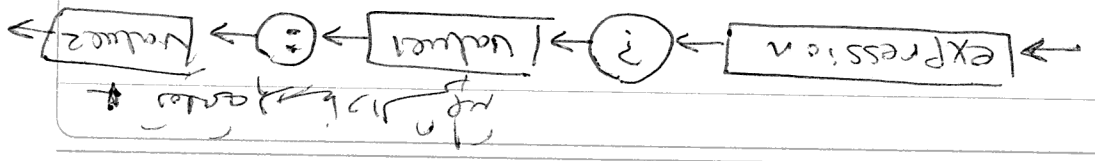
if expression:
instructionBlock

else:
instructionBlock

if-elif ...

if expression:
instructionBlock

elif expression:
instructionBlock



9 *عندما نعطى قيمة لـ expression*

Value 2
Value 1

11 *A = value 1 if expression else value 2*

12 *x = -15*
absx = x if (x >= 0) else -x
print(absx)

1 *x = not(a)*

2 *while*
while expression:
instructionBlock

4 *max = input("a = ")*
a = int(a)
max = a

5 *while (a != 999):*
if (a > max):

6 *max = a*
a = input("a = ")
a = int(a)
if (max != 999):

7 *print("max = ", max)*

8 *تسليم البرنامج الى المستخدم*

Thursday 23

Jendredi

June - Juin

٢٣ جمادى الأولى

حزيران - يونيو

24 DhulQadah 11/1443

Day: 174 / 191

2022

Week 25, Month 6

٢٤ ذو القعدة ١٤٤٣/١١

For $i = 1$ to N step 1
 $M = i^2 + 3i + 1$
 Print M

10 #Read M
 Print("input M:");
 SM = input(); M = int(SM)
 # Clear S
 S = 0

11 # compute the sum in S
 For i in range(1, M+1):
 S = S + i;
 # write S
 Print("1+2+...+M, n=","S");

3 $M = 1 \times 2 \times 3 \times \dots \times N$
4 $M = 1 \times 2 \times 3 \times \dots \times N$
5 $M = 1 \times 2 \times 3 \times \dots \times N$
6 $M = 1 \times 2 \times 3 \times \dots \times N$
7 $M = 1 \times 2 \times 3 \times \dots \times N$
8 $M = 1 \times 2 \times 3 \times \dots \times N$

#Read M
 Print("input M:");
 SM = input(); M = int(SM);
 # Clear S
 S = 1
 # compute the M! in S

22

uṇṇ

۱۰۰

۱۰۱ - حنین

[illegible]

Week 25, Month 6

2022

Day: 173 / 192

23 DhuIQadāh 11/1443

For i in range $(1, M+1)$:

$$S = S^*$$

#write 8

```
print("w' u' s)");
```

17/11/2017 break + يوم الغرة *
الغرة يوم الغرة

11/2

$$1 = 0! \quad n = 13$$

```
while (i < n):
```

$$f: ((n \leq 1) \text{ and } (0 = n \wedge 1)) \rightarrow 1$$

break

```
print(u'!')
```

$$1+1=2$$
[illegible]

15/11/2023 12:00 PM

3. 11. 1963 4. 11. 1963

الحمد لله الذي جعلنا من عباده الصالحين

Ullmann & Ullmann

1. What is the purpose of the experiment? → Continue area *

23. 11. 2017

Break Fast is a 2nd grade story.

1. $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$
 2. $\frac{1}{2} \times \frac{1}{4} = \frac{1}{8}$
 3. $\frac{1}{4} \times \frac{1}{4} = \frac{1}{16}$
 4. $\frac{1}{2} \times \frac{1}{8} = \frac{1}{16}$
 5. $\frac{1}{4} \times \frac{1}{8} = \frac{1}{32}$
 6. $\frac{1}{8} \times \frac{1}{8} = \frac{1}{64}$
 7. $\frac{1}{2} \times \frac{1}{16} = \frac{1}{32}$
 8. $\frac{1}{4} \times \frac{1}{16} = \frac{1}{64}$
 9. $\frac{1}{8} \times \frac{1}{16} = \frac{1}{128}$
 10. $\frac{1}{2} \times \frac{1}{32} = \frac{1}{64}$
 11. $\frac{1}{4} \times \frac{1}{32} = \frac{1}{128}$
 12. $\frac{1}{8} \times \frac{1}{32} = \frac{1}{256}$
 13. $\frac{1}{2} \times \frac{1}{64} = \frac{1}{32}$
 14. $\frac{1}{4} \times \frac{1}{64} = \frac{1}{256}$
 15. $\frac{1}{8} \times \frac{1}{64} = \frac{1}{512}$
 16. $\frac{1}{2} \times \frac{1}{128} = \frac{1}{64}$
 17. $\frac{1}{4} \times \frac{1}{128} = \frac{1}{512}$
 18. $\frac{1}{8} \times \frac{1}{128} = \frac{1}{1024}$
 19. $\frac{1}{2} \times \frac{1}{256} = \frac{1}{128}$
 20. $\frac{1}{4} \times \frac{1}{256} = \frac{1}{1024}$
 21. $\frac{1}{8} \times \frac{1}{256} = \frac{1}{2048}$
 22. $\frac{1}{2} \times \frac{1}{512} = \frac{1}{256}$
 23. $\frac{1}{4} \times \frac{1}{512} = \frac{1}{2048}$
 24. $\frac{1}{8} \times \frac{1}{512} = \frac{1}{4096}$
 25. $\frac{1}{2} \times \frac{1}{1024} = \frac{1}{512}$
 26. $\frac{1}{4} \times \frac{1}{1024} = \frac{1}{4096}$
 27. $\frac{1}{8} \times \frac{1}{1024} = \frac{1}{8192}$
 28. $\frac{1}{2} \times \frac{1}{2048} = \frac{1}{1024}$
 29. $\frac{1}{4} \times \frac{1}{2048} = \frac{1}{8192}$
 30. $\frac{1}{8} \times \frac{1}{2048} = \frac{1}{16384}$
 31. $\frac{1}{2} \times \frac{1}{4096} = \frac{1}{2048}$
 32. $\frac{1}{4} \times \frac{1}{4096} = \frac{1}{16384}$
 33. $\frac{1}{8} \times \frac{1}{4096} = \frac{1}{32768}$
 34. $\frac{1}{2} \times \frac{1}{8192} = \frac{1}{4096}$
 35. $\frac{1}{4} \times \frac{1}{8192} = \frac{1}{32768}$
 36. $\frac{1}{8} \times \frac{1}{8192} = \frac{1}{65536}$
 37. $\frac{1}{2} \times \frac{1}{16384} = \frac{1}{8192}$
 38. $\frac{1}{4} \times \frac{1}{16384} = \frac{1}{65536}$
 39. $\frac{1}{8} \times \frac{1}{16384} = \frac{1}{131072}$
 40. $\frac{1}{2} \times \frac{1}{32768} = \frac{1}{16384}$
 41. $\frac{1}{4} \times \frac{1}{32768} = \frac{1}{131072}$
 42. $\frac{1}{8} \times \frac{1}{32768} = \frac{1}{262144}$
 43. $\frac{1}{2} \times \frac{1}{65536} = \frac{1}{32768}$
 44. $\frac{1}{4} \times \frac{1}{65536} = \frac{1}{262144}$
 45. $\frac{1}{8} \times \frac{1}{65536} = \frac{1}{524288}$
 46. $\frac{1}{2} \times \frac{1}{131072} = \frac{1}{65536}$
 47. $\frac{1}{4} \times \frac{1}{131072} = \frac{1}{524288}$
 48. $\frac{1}{8} \times \frac{1}{131072} = \frac{1}{1048576}$
 49. $\frac{1}{2} \times \frac{1}{262144} = \frac{1}{131072}$
 50. $\frac{1}{4} \times \frac{1}{262144} = \frac{1}{1048576}$
 51. $\frac{1}{8} \times \frac{1}{262144} = \frac{1}{2097152}$
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 65. $\frac{1}{4} \times \frac{1}{8388608} = \frac{1}{33554432}$
 66. $\frac{1}{8} \times \frac{1}{8388608} = \frac{1}{67108864}$
 67. $\frac{1}{2} \times \frac{1}{16777216} = \frac{1}{8388608}$
 68. $\frac{1}{4} \times \frac{1}{16777216} = \frac{1}{67108864}$
 69. $\frac{1}{8} \times \frac{1}{16777216} = \frac{1}{134217728}$
 70. $\frac{1}{2} \times \frac{1}{33554432} = \frac{1}{16777216}$
 71. $\frac{1}{4} \times \frac{1}{33554432} = \frac{1}{134217728}$
 72. $\frac{1}{8} \times \frac{1}{33554432} = \frac{1}{268435456}$
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 74. $\frac{1}{4} \times \frac{1}{67108864} = \frac{1}{268435456}$
 75. $\frac{1}{8} \times \frac{1}{67108864} = \frac{1}{536870912}$
 76. $\frac{1}{2} \times \frac{1}{134217728} = \frac{1}{67108864}$
 77. $\frac{1}{4} \times \frac{1}{134217728} = \frac{1}{536870912}$
 78. $\frac{1}{8} \times \frac{1}{134217728} = \frac{1}{1073741824}$
 79. $\frac{1}{2} \times \frac{1}{268435456} = \frac{1}{134217728}$
 80. $\frac{1}{4} \times \frac{1}{268435456} = \frac{1}{1073741824}$
 81. $\frac{1}{8} \times \frac{1}{268435456} = \frac{1}{2147483648}$
 82. $\frac{1}{2} \times \frac{1}{536870912} = \frac{1}{268435456}$
 83. $\frac{1}{4} \times \frac{1}{536870912} = \frac{1}{2147483648}$
 84. $\frac{1}{8} \times \frac{1}{536870912} = \frac{1}{4294967296}$
 85. $\frac{1}{2} \times \frac{1}{1073741824} = \frac{1}{536870912}$
 86. $\frac{1}{4} \times \frac{1}{1073741824} = \frac{1}{4294967296}$
 87. $\frac{1}{8} \times \frac{1}{1073741824} = \frac{1}{8589934592}$
 88. $\frac{1}{2} \times \frac{1}{2147483648} = \frac{1}{1073741824}$
 89. $\frac{1}{4} \times \frac{1}{2147483648} = \frac{1}{8589934592}$
 90. $\frac{1}{8} \times \frac{1}{2147483648} = \frac{1}{17179869184}$
 91. $\frac{1}{2} \$

مجلسه اول - ۱۳۹۸

1. התאחדות העבודה - התאחדות העובדים

Tuesday
Mardi

June - Juin

٢١
الجمعة

حزيران - يونيو

٢٢ ذو القعدة ١٤٤٣/١١

Week 25, Month 6

2022

Day: 172 / 193

22 Dhu'l-Qadah 11/1443

For i in range(0,13+1):
if ((i%4==0) and (i>4)):
continue!
Print("i = ",i)!

1 = 1
2 = 2
3 = 3
4 = 4
5 = 5
6 = 6
7 = 7
8 = 8
9 = 9
10 = 10
11 = 11
12 = 12
13 = 13

الجمعة ٢٢ ذو القعدة ١٤٤٣/١١

n=0
sum=0
#Read n
n=int(input("input n:"))
while(n!=999):
sum=sum+n
n=int(input("input n:"))
Print("sum = ",sum)

JUN
9

int float, complex, string, bool, set, list, tuple, dict

(dict, list, tuple, set, list, tuple, dict)

الانواع

assign Integer

i = 34

نوع

assign float

f = 105.7

assign complex

c = 5+4j

assign Boolean

b = True

assign string

s = "This is a string"

assign Integer

a = 34

assign float

a = 105.7

assign Boolean

a = True

assign string

a = "This is a string"

assign string

a = "This is a string"

assign string

a = "This is a string"

June - July

کتابت = کتاب

Day: 170 / 195

2022

Week 25, Month 6

1675/11 11/12/1950

Float 2 1/2 s

Complex good ss

absolutely not, round (1) 17/12/2017

* $\text{round}(\frac{\text{sum}(m_i)}{n})$, $m_i \in \mathbb{N}$

Sum (10, 15, 3) = 28

$$\text{sum}(15) = 15$$

True or False: $\lim_{x \rightarrow 0} \frac{1}{x} = 0$ +

and, or, not model
truth table model

⇒

NR
9

#testing the conditional AND operators (ϕϕ)

Print("conditional AND (and)" +

"\nfalse and false:" + str(False and False) +

"\nfalse and true:" + str(False and True) +

"\ntrue and false:" + str(True and False) +

"\ntrue and true:" + str(True and True) +

#testing the conditional OR operators (11)

Print("\nconditional OR (or)" +

"\nfalse or false:" + str(False or False) +

"\nfalse or true:" + str(False or True) +

"\ntrue or false:" + str(True or False) +

"\ntrue or true:" + str(True or True) +

Logical AND (ϕ)

Logical OR (1)

Logical Exclusive OR (1)

Logical NOT (not)

Conditional AND (and)

True	True	True
True	False	False
False	True	False
False	False	True

Conditional OR (or)

True	True	True
True	False	True
False	True	True
False	False	False

Logical AND (and)

True	True	True
True	False	False
False	True	False
False	False	False

Logical OR (or)

True	True	True
True	False	True
False	True	True
False	False	False

Logical Exclusive OR (xor)

True	True	False
True	False	True
False	True	True
False	False	False

Logical NOT (not)

not False: True
not True: False

Bitwise operation

operation

$$1 \text{ } \dot{\gamma} = \gamma \text{ } \dot{\gamma}$$
$$10000010 = 65$$

35 = 00100011

10. $\frac{1}{x^2} = x^{-2}$ $\frac{d}{dx} x^{-2} = -2x^{-3} = -\frac{2}{x^3}$

next) $\frac{1}{2}$

$$35 = 70$$

```
print(bin(a)) ⇒ 0b100011
```

$\text{print}(\text{oct}(a)) \Rightarrow 0043$

print(hex(a)) $\Rightarrow 0x23$

A handwritten signature consisting of stylized letters 'G', 'H', and 'B' written across horizontal lines. The 'G' is large and loops around the 'H'. The 'B' is also large and loops around the 'H'. There are some additional scribbles below the main letters.

1	1	1
0	0	0
1	0	0

(7th)

1	1	1
1	0	0
1	0	0

A handwritten grid on lined paper. The grid consists of three rows and three columns. The numbers in the grid are as follows:

0	1	1
1	0	0
1	0	104

The number '104' in the bottom-right cell is circled and has a small arrow pointing to it from above.

$$86 \div 01000110 = 98 \vee 99$$
$$66 = 1000110 = 5E | 59$$
$$65 \div 35 = 1.8571428571428571$$

59 = 70

58 = 9

$\text{Print}(a, n \text{ of } n, b, n = n, a \& b) \Rightarrow 65 \text{ of } 35 = 1$
 $\text{Print}(a, n \text{ of } n, b, n = n, a \mid b) \Rightarrow 65 \mid 35 = 99$
 $\text{Print}(a, n \text{ of } n, b, n = n, a \vee b) \Rightarrow 65 \vee 35 = 98$

10 $F_1 = \text{input}(\text{"Function 1: } F_1(x) = \text{"})$

```
print ("F1(x)", F1)
print ("F2(x)", F2)
```

12 $\Delta x = 0,25$

For x in range $(0, 10)$

$$y_1 = \cosh\left(\frac{x}{a}\right)$$
$$y_2 = \text{eval}(f_2)$$

```
print(" %.2f" % x, y, x, y)
```

$\Rightarrow$ Function 1: $f_1(x) = x + 2$
 Function 2: $f_2(x) = \text{pow}(x, 3)$

F1C11 X#A-2

$$F_2(X) \quad \text{pow}(X, 3)$$

00-0 00-0 00-0

6
0.25

0.50 0.25

0.75 7

00-1 00-1

8

1.50

13 الثلاثاء
حزيران - يونيو

10 ذو القعدة 1443/11

Week 24, Month 6

2022

Day: 165 / 200

15 DhulQadah 11/1443

June - Juin

Tuesday
Mardi

14

5.36	3.06	1.75
8.00	4.00	2.00
11.39	5.06	2.25

ما قبله من ك ما بعده من ك
input k in [2..10] = 5

5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50

print("input k in [2..10] = ")
k = int(input())
ك ما قبله من ك ما بعده من ك

while (i <= 10):
print(k, "x", i, "=", k*i)
i = i + 1

Monday
Lundi
13

June - Juin

14 Dhulqadad 11/1443

Day: 164 / 201

2022

Week 24, Month 6

13 ذو القعدة 1443/11

حزيران - يونيو

الاثنين
13

8 $ax^2 + bx + c = 0$
 $\Delta = b^2 - 4ac$
طريقة الجذور التربيعية
($a \neq 0$)

9 import math
a, b, c are
10 sa = input("input a = ")
11 sb = input("input b = ")
12 sc = input("input c = ")
a = float(sa)
b = float(sb)
c = float(sc)

12 $\Delta = b^2 - 4 \times a \times c$
الجواب

2 if ($\Delta < 0$):
Print("No Real solutions")
if ($\Delta == 0$):
Print("one solution")

4 $x = -b / (2 \times a)$
Print("x = " + str(x))
if ($\Delta > 0$):
Print("two solutions")

6 $x1 = (-b - \text{math.sqrt}(\Delta)) / (2 \times a)$
 $x2 = (-b + \text{math.sqrt}(\Delta)) / (2 \times a)$
7 Print("x1 = ", x1, " x2 = ", x2)

12

D:manche

June - July

13 DhuIQadah 11/1443

Day: 163 / 202

2022

Week 24, Month 6

۱۱ / ۲۳۶ ۵۰۴۸۱۱ نو ۱۱

وقت - زبان

Handwritten cursive letters 'א' and 'ב' with stroke order arrows. 'א' is formed by a downward stroke on the left and a curved stroke on the right. 'ב' is formed by a single downward stroke.

$a^2 + b^2 = c^2$

$$a \cdot x^2 + b \cdot x + c = 0 \quad \text{--- (1)}$$

import math

9/10/25

$$sa = \text{input}('input - a = '); a = \text{float}(sa)$$
$$sb = \text{input}('input b = '); \quad b = \text{float}(sb);$$
$$:(\alpha = \beta) \text{ f!}$$
$$\frac{1}{2}(0 \rightarrow 9) \text{ 个}$$
$$\delta(0 \equiv 2) - \delta!$$

```
Print ("Each real is a solution")
else: # c1=0
```

```
print("incorrect equation")
```

$$X = -C/D$$

```
else, # a[i] = 0
    print("The solution is : " + str(x))
```

$$\Delta a = b \times d - c \times a = 14 \times 9 - 7 \times 9 = 63$$
$$IF(\alpha \leq 0) = 1$$

if (delta == 0):

Print (none)

$$x = -b/2a$$
$$f: \mathbb{R} \rightarrow \mathbb{R} \text{ (decreasing)}$$

Saturday 11

Samedi

June - Juin

12 DhulQadah 11/1443

Day: 162 / 203

Week 24, Month 6

2022

15/11/2022

جوان - يونيو

السنين

Print - ("Two Solutions")
 $x_1 = (-b - \text{math.sqrt}(\text{delta})) / (2 * a)$
 $x_2 = (-b + \text{math.sqrt}(\text{delta})) / (2 * a)$
 $\text{print}(x_1 = " + \text{str}(x_1) + " + \text{str}(x_2) + "$

Perfect - "Perfect"

1 + 2 + 3 = 6

perfect - "perfect"

1 $\text{compt} = 0$
2 $\text{print}(\text{"Number of perfect numbers you wish find"})$
3 $n = \text{int}(\text{input}())$
4 number

5 $\text{while compt} < n$
6 $\text{nbr} = 2$

7 $\text{while nbr} <= n / 2$
8 $\text{if nbr \% k} == 0$

9 $\text{if (sumdiv == nbr)}$
10 $\text{print(str(nbr) + " is a perfect number")}$
11 $\text{compt} += 1$
12 $\text{nbr} += 1$

[illegible]

English Keyword	Arabic Translation
1 and	و
2 as	كما
3 assert	تأكيد
4 break	كسر
5 class	فئة
6 continue	استمرار
7 def	تعريف
8 del	حذف
9 elif	إلا إذا
10 else	إلا
11 except	إلا
12 False	خاطئ
13 finally	في النهاية
14 for	لـ
15 from	من
16 Global	عالمي
17 if	إذا
18 import	استيراد
19 in	في
20 is	هو
21 lambda	دالة لامدا

(27) None
(28) nonlocal

(24) not

(25) or

(26) pass

(27) raise

(28) return

(29) True

(30) try

(31) while

(32) with

(33) yield

JUN 9

تجربة تايپ null
تجربة تايپ non-local variable
تجربة تايپ null, pass, return, yield, while, try, with, raise, not, or, pass, return, True, try, while, with, yield

تجربة تايپ raise exception
تجربة تايپ raise exception

تجربة تايپ while
تجربة تايپ while

تجربة تايپ Built-in Functions
تجربة تايپ Built-in Functions

3 abs(), bin(), bool(), complex(), eval(), float(),
format(), hex(), input(), int(), len(), max(),
4 min(), oct(), ord(), pow(), print(), range(), repr(),
round(), str(), sum(), type()

تجربة تايپ *
تجربة تايپ *
تجربة تايپ *

format()	→	تنسيق
freezeSet()	→	جمد مجموعة
getAttr()	→	الحصول على خاصية (التي هي property)
globals()	→	الحصول على قائمة بالمتغيرات الجارية في النطاق العالمي (global symbol table)
hasAttr()	→	تحقق من وجود خاصية (property) في كائن ما
hash()	→	حساب قيمة hash
help()	→	تقديم نصائح حول استخدام built-in
hex()	→	تحويل القيمة إلى شكل سداسي عشري (hexadecimal)
id()	→	تقديم معرف للقيمة
input()	→	الحصول على مدخل من المستخدم
int()	→	تحويل القيمة إلى نوع Integer
instance()	→	تحقق من أن الكائن هو مثيل لـ class
isSubclass()	→	تحقق من أن الكلاس هو فرع لـ subclass
iter()	→	تقديم كائن iter-object
len()	→	تقديم طول الكائن
list()	→	تقديم قائمة
locals()	→	الحصول على قائمة بالمتغيرات المحلية (local symbol table)
map()	→	تطبيق دالة على كل عنصر في القائمة
max()	→	تقديم أكبر قيمة

9

کتابخانه



2203

Week23, Month6

1631/11 10/10/1956

• ایک (۱) (۲) (۳) (۴) (۵) (۶) (۷) (۸) (۹) (۱۰) (۱۱) (۱۲) (۱۳) (۱۴) (۱۵) (۱۶) (۱۷) (۱۸) (۱۹) (۲۰) (۲۱) (۲۲) (۲۳) (۲۴) (۲۵) (۲۶) (۲۷) (۲۸) (۲۹) (۳۰) (۳۱) (۳۲) (۳۳) (۳۴) (۳۵) (۳۶) (۳۷) (۳۸) (۳۹) (۴۰) (۴۱) (۴۲) (۴۳) (۴۴) (۴۵) (۴۶) (۴۷) (۴۸) (۴۹) (۵۰) (۵۱) (۵۲) (۵۳) (۵۴) (۵۵) (۵۶) (۵۷) (۵۸) (۵۹) (۶۰) (۶۱) (۶۲) (۶۳) (۶۴) (۶۵) (۶۶) (۶۷) (۶۸) (۶۹) (۷۰) (۷۱) (۷۲) (۷۳) (۷۴) (۷۵) (۷۶) (۷۷) (۷۸) (۷۹) (۸۰) (۸۱) (۸۲) (۸۳) (۸۴) (۸۵) (۸۶) (۸۷) (۸۸) (۸۹) (۹۰) (۹۱) (۹۲) (۹۳) (۹۴) (۹۵) (۹۶) (۹۷) (۹۸) (۹۹) (۱۰۰) (۱۰۱) (۱۰۲) (۱۰۳) (۱۰۴) (۱۰۵) (۱۰۶) (۱۰۷) (۱۰۸) (۱۰۹) (۱۱۰) (۱۱۱) (۱۱۲) (۱۱۳) (۱۱۴) (۱۱۵) (۱۱۶) (۱۱۷) (۱۱۸) (۱۱۹) (۱۲۰) (۱۲۱) (۱۲۲) (۱۲۳) (۱۲۴) (۱۲۵) (۱۲۶) (۱۲۷) (۱۲۸) (۱۲۹) (۱۳۰) (۱۳۱) (۱۳۲) (۱۳۳) (۱۳۴) (۱۳۵) (۱۳۶) (۱۳۷) (۱۳۸) (۱۳۹) (۱۴۰) (۱۴۱) (۱۴۲) (۱۴۳) (۱۴۴) (۱۴۵) (۱۴۶) (۱۴۷) (۱۴۸) (۱۴۹) (۱۵۰) (۱۵۱) (۱۵۲) (۱۵۳) (۱۵۴) (۱۵۵) (۱۵۶) (۱۵۷) (۱۵۸) (۱۵۹) (۱۶۰) (۱۶۱) (۱۶۲) (۱۶۳) (۱۶۴) (۱۶۵) (۱۶۶) (۱۶۷) (۱۶۸) (۱۶۹) (۱۷۰) (۱۷۱) (۱۷۲) (۱۷۳) (۱۷۴) (۱۷۵) (۱۷۶) (۱۷۷) (۱۷۸) (۱۷۹) (۱۸۰) (۱۸۱) (۱۸۲) (۱۸۳) (۱۸۴) (۱۸۵) (۱۸۶) (۱۸۷) (۱۸۸) (۱۸۹) (۱۹۰) (۱۹۱) (۱۹۲) (۱۹۳) (۱۹۴) (۱۹۵) (۱۹۶) (۱۹۷) (۱۹۸) (۱۹۹) (۲۰۰) (۲۰۱) (۲۰۲) (۲۰۳) (۲۰۴) (۲۰۵) (۲۰۶) (۲۰۷) (۲۰۸) (۲۰۹) (۲۱۰) (۲۱۱) (۲۱۲) (۲۱۳) (۲۱۴) (۲۱۵) (۲۱۶) (۲۱۷) (۲۱۸) (۲۱۹) (۲۲۰) (۲۲۱) (۲۲۲) (۲۲۳) (۲۲۴) (۲۲۵) (۲۲۶) (۲۲۷) (۲۲۸) (۲۲۹) (۲۳۰) (۲۳۱) (۲۳۲) (۲۳۳) (۲۳۴) (۲۳۵) (۲۳۶) (۲۳۷) (۲۳۸) (۲۳۹) (۲۴۰) (۲۴۱) (۲۴۲) (۲۴۳) (۲۴۴) (۲۴۵) (۲۴۶) (۲۴۷) (۲۴۸) (۲۴۹) (۲۵۰) (۲۵۱) (۲۵۲) (۲۵۳) (۲۵۴) (۲۵۵) (۲۵۶) (۲۵۷) (۲۵۸) (۲۵۹) (۲۶۰) (۲۶۱) (۲۶۲) (۲۶۳) (۲۶۴) (۲۶۵) (۲۶۶) (۲۶۷) (۲۶۸) (۲۶۹) (۲۷۰) (۲۷۱) (۲۷۲) (۲۷۳) (۲۷۴) (۲۷۵) (۲۷۶) (۲۷۷) (۲۷۸) (۲۷۹) (۲۸۰) (۲۸۱) (۲۸۲) (۲۸۳) (۲۸۴) (۲۸۵) (۲۸۶) (۲۸۷) (۲۸۸) (۲۸۹) (۲۹۰) (۲۹۱) (۲۹۲) (۲۹۳) (۲۹۴) (۲۹۵) (۲۹۶) (۲۹۷) (۲۹۸) (۲۹۹) (۳۰۰) (۳۰۱) (۳۰۲) (۳۰۳) (۳۰۴) (۳۰۵) (۳۰۶) (۳۰۷) (۳۰۸) (۳۰۹) (۳۱۰) (۳۱۱) (۳۱۲) (۳۱۳) (۳۱۴) (۳۱۵) (۳۱۶) (۳۱۷) (۳۱۸) (۳۱۹) (۳۲۰) (۳۲۱) (۳۲۲) (۳۲۳) (۳۲۴) (۳۲۵) (۳۲۶) (۳۲۷) (۳۲۸) (۳۲۹) (۳۳۰) (۳۳۱) (۳۳۲) (۳۳۳) (۳۳۴) (۳۳۵) (۳۳۶) (۳۳۷) (۳۳۸) (۳۳۹) (۳۴۰) (۳۴۱) (۳۴۲) (۳۴۳) (۳۴۴) (۳۴۵) (۳۴۶) (۳۴۷) (۳۴۸) (۳۴۹) (۳۵۰) (۳۵۱) (۳۵۲) (۳۵۳) (۳۵۴) (۳۵۵) (۳۵۶) (۳۵۷) (۳۵۸) (۳۵۹) (۳۶۰) (۳۶۱) (۳۶۲) (۳۶۳) (۳۶۴) (۳۶۵) (۳۶۶) (۳۶۷) (۳۶۸) (۳۶۹) (۳۷۰) (۳۷۱) (۳۷۲) (۳۷۳) (۳۷۴) (۳۷۵) (۳۷۶) (۳۷۷) (۳۷۸) (۳۷۹) (۳۸۰) (۳۸۱) (۳۸۲) (۳۸۳) (۳۸۴) (۳۸۵) (۳۸۶) (۳۸۷) (۳۸۸) (۳۸۹) (۳۹۰) (۳۹۱) (۳۹۲) (۳۹۳) (۳۹۴) (۳۹۵) (۳۹۶) (۳۹۷) (۳۹۸) (۳۹۹) (۴۰۰) (۴۰۱) (۴۰۲) (۴۰۳) (۴۰۴) (۴۰۵) (۴۰۶) (۴۰۷) (۴۰۸) (۴۰۹) (۴۱۰) (۴۱۱) (۴۱۲) (۴۱۳) (۴۱۴) (۴۱۵) (۴۱۶) (۴۱۷) (۴۱۸) (۴۱۹) (۴۲۰) (۴۲۱) (۴۲۲) (۴۲۳) (۴۲۴) (۴۲۵) (۴۲۶) (۴۲۷) (۴۲۸) (۴۲۹) (۴۳۰) (۴۳۱) (۴۳۲) (۴۳۳) (۴۳۴) (۴۳۵) (۴۳۶) (۴۳۷) (۴۳۸) (۴۳۹) (۴۴۰) (۴۴۱) (۴۴۲) (۴۴۳) (۴۴۴) (۴۴۵) (۴۴۶) (۴۴۷) (۴۴۸) (۴۴۹) (۴۵۰) (۴۵۱) (۴۵۲) (۴۵۳) (۴۵۴) (۴۵۵) (۴۵۶) (۴۵۷) (۴۵۸) (۴۵۹) (۴۶۰) (۴۶۱) (۴۶۲) (۴۶۳) (۴۶۴) (۴۶۵) (۴۶۶) (۴۶۷) (۴۶۸) (۴۶۹) (۴۷۰) (۴۷۱) (۴۷۲) (۴۷۳) (۴۷۴) (۴۷۵) (۴۷۶) (۴۷۷) (۴۷۸) (۴۷۹) (۴۸۰) (۴۸۱) (۴۸۲) (۴۸۳) (۴۸۴) (۴۸۵) (۴۸۶) (۴۸۷) (۴۸۸) (۴۸۹) (۴۹۰) (۴۹۱) (۴۹۲) (۴۹۳) (۴۹۴) (۴۹۵) (۴۹۶) (۴۹۷) (۴۹۸) (۴۹۹) (۵۰۰) (۵۰۱) (۵۰۲) (۵۰۳) (۵۰۴) (۵۰۵) (۵۰۶) (۵۰۷) (۵۰۸) (۵۰۹) (۵۱۰) (۵۱۱) (۵۱۲) (۵۱۳) (۵۱۴) (۵۱۵) (۵۱۶) (۵۱۷) (۵۱۸) (۵۱۹) (۵۲۰) (۵۲۱) (۵۲۲) (۵۲۳) (۵۲۴) (۵۲۵) (۵۲۶) (۵۲۷) (۵۲۸) (۵۲۹) (۵۳۰) (۵۳۱) (۵۳۲) (۵۳۳) (۵۳۴) (۵۳۵) (۵۳۶) (۵۳۷) (۵۳۸)

1. वैदिक काल

Hemable $C_1H_{\frac{1}{2}}O_{\frac{1}{2}}$

Object - no real

[illegible]

File object

Unit Code DT 07

Y = 0.0001 + 0.0001X

17. 11. 2019

ॐ नमो भगवते वासुदेवाय

gete set

1. الزكاة (Zakat)

1. अभिज्ञान शाकुन्तलम्

10/10/2020

[illegible]

$\frac{d}{dt} \left(\frac{1}{2} m v^2 \right) = \frac{1}{2} m \frac{d}{dt} (v^2)$

[illegible]

1871

55 P 17 000000 17 24 45

FAIR - POTHEN

Strong and well

[illegible]

Items

جاء في كتابه

JUN 6

tuple() → tuple

type() → نوع النوع

vars() → متغيرات

zip() → ضغط

Array - List - المصفوفات - القائمة

المصفوفة - المصفوفة

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**Saturday
Samed!**

June - July

5 Phulbadi 11/1443

Day: 155 / 210

2202

Week23, Month6

11/2331 11/2331

$se = \text{input}(\text{"voice for: "})$
 $e = \text{int}(se)$

$$e = \inf(S)$$

~~# write results ce~~

```
print("Candidate 1 = ", c1)
print("Candidate 2 = ", c2)
```

Communicate $z =$, $c(2)$

Prin- (n Cand: $\text{dot} \cdot \text{dot} = \text{dot} \cdot \text{dot}$)

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$$C = [0] \neq [0] \quad (1, 0, 1)$$

$se = \text{input}(\text{"Rate for : "})$
 $se = \text{int}(se)$

```
while e != 0:
```

cc J. L.

if $e \geq 1$ and $e < 4$:

 ~~$C[0] \neq 1$~~

کتبہ - ج ۱۱۲

١٢

```

Se=input("Vote for: ")

```

$$e = \ln t(\text{se})$$

6 כבש קרנול

for c in range $(1, nc+1)$:

e in range $(1, nC+1)$;
Print("Candidate" + str(e) + " = " + str(c[e]))

12. $\text{type}(c) \rightarrow \text{list}$
 13. NC

list $\leftarrow$ type(c)

$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ -1 & i \end{pmatrix}$

1. What is the main purpose of the study?
 2. What are the research objectives?
 3. What is the significance of the study?
 4. What are the limitations of the study?
 5. What are the conclusions of the study?

2. 5. 7. 9. 11. 13. 15. 17. 19. 21. 23. 25. 27. 29. 31. 33. 35. 37. 39. 41. 43. 45. 47. 49. 51. 53. 55. 57. 59. 61. 63. 65. 67. 69. 71. 73. 75. 77. 79. 81. 83. 85. 87. 89. 91. 93. 95. 97. 99. 101. 103. 105. 107. 109. 111. 113. 115. 117. 119. 121. 123. 125. 127. 129. 131. 133. 135. 137. 139. 141. 143. 145. 147. 149. 151. 153. 155. 157. 159. 161. 163. 165. 167. 169. 171. 173. 175. 177. 179. 181. 183. 185. 187. 189. 191. 193. 195. 197. 199. 201. 203. 205. 207. 209. 211. 213. 215. 217. 219. 221. 223. 225. 227. 229. 231. 233. 235. 237. 239. 241. 243. 245. 247. 249. 251. 253. 255. 257. 259. 261. 263. 265. 267. 269. 271. 273. 275. 277. 279. 281. 283. 285. 287. 289. 291. 293. 295. 297. 299. 301. 303. 305. 307. 309. 311. 313. 315. 317. 319. 321. 323. 325. 327. 329. 331. 333. 335. 337. 339. 341. 343. 345. 347. 349. 351. 353. 355. 357. 359. 361. 363. 365. 367. 369. 371. 373. 375. 377. 379. 381. 383. 385. 387. 389. 391. 393. 395. 397. 399. 401. 403. 405. 407. 409. 411. 413. 415. 417. 419. 421. 423. 425. 427. 429. 431. 433. 435. 437. 439. 441. 443. 445. 447. 449. 451. 453. 455. 457. 459. 461. 463. 465. 467. 469. 471. 473. 475. 477. 479. 481. 483. 485. 487. 489. 491. 493. 495. 497. 499. 501. 503. 505. 507. 509. 511. 513. 515. 517. 519. 521. 523. 525. 527. 529. 531. 533. 535. 537. 539. 541. 543. 545. 547. 549. 551. 553. 555. 557. 559. 561. 563. 565. 567. 569. 571. 573. 575. 577. 579. 581. 583. 585. 587. 589. 591. 593. 595. 597. 599. 601. 603. 605. 607. 609. 611. 613. 615. 617. 619. 621. 623. 625. 627. 629. 631. 633. 635. 637. 639. 641. 643. 645. 647. 649. 651. 653. 655. 657. 659. 661. 663. 665. 667. 669. 671. 673. 675. 677. 679. 681. 683. 685. 687. 689. 691. 693. 695. 697. 699. 701. 703. 705. 707. 709. 711. 713. 715. 717. 719. 721. 723. 725. 727. 729. 731. 733. 735. 737. 739. 741. 743. 745. 747. 749. 751. 753. 755. 757. 759. 761. 763. 765. 767. 769. 771. 773. 775. 777. 779. 781. 783. 785. 787. 789. 791. 793. 795. 797. 799. 801. 803. 805. 807. 809. 811. 813. 815. 817. 819. 821. 823. 825. 827. 829. 831. 833. 835. 837. 839. 841. 843. 845. 847. 849. 851. 853. 855. 857. 859. 861. 863. 865. 867. 869. 871. 873. 875. 877. 879. 881. 883. 885. 887. 889. 891. 893. 895. 897. 899. 901. 903. 905. 907. 909. 911. 913. 915. 917. 919. 921. 923. 925. 927. 929. 931. 933. 935. 937. 939. 941. 943. 945. 947. 949. 951. 953. 955. 957. 959. 961. 963. 965. 967. 969. 971. 973. 975. 977. 979. 981. 983. 985. 987. 989. 991. 993. 995. 997. 999. 1001. 1003. 1005. 1007. 1009. 1011. 1013. 1015. 1017. 1019. 1021. 1023. 1025. 1027. 1029. 1031. 1033. 1035. 1037. 1039. 1041. 1043. 1045. 1047. 1049. 1051. 1053. 1055. 1057. 1059. 1061. 1063. 1065. 1067. 1069. 1071. 1073. 1075. 1077. 1079. 1081. 1083. 1085. 1087. 1089. 1091. 1093. 1095. 1097. 1099. 1101. 1103. 1105. 1107. 1109. 1111. 1113. 1115. 1117. 1119. 1121. 1123. 1125. 1127. 1129. 1131. 1133. 1135. 1137. 1139. 1141. 1143. 1145. 1147. 1149. 1151. 1153. 1155. 1157. 1159. 1161. 1163. 1165. 1167. 1169. 1171. 1173. 1175. 1177. 1179. 1181. 1183. 1185. 1187. 1189. 1191. 1193. 1195. 1197. 1199. 1201. 1203. 1205. 1207. 1209. 1211. 1213. 1215. 1217. 1219. 1221. 1223. 1225. 1227. 1229. 1231. 1233. 1235. 1237. 1239. 1241. 1243. 1245. 1247. 1249. 1251. 1253. 1255. 1257. 1259. 1261. 1263. 1265. 1267. 1269. 1271. 1273. 1275. 1277. 1279. 1281. 1283. 1285. 1287. 1289. 1291. 1293. 1295. 1297. 1299. 1301. 1303. 1305. 1307. 1309. 1311. 1313. 1315. 1317. 1319. 1321. 1323. 1325. 1327. 1329. 1331. 1333. 1335. 1337. 1339. 1341. 1343. 1345. 1347. 1349. 1351. 1353. 1355. 1357. 1359. 1361. 1363. 1365. 1367. 1369. 1371. 1373. 1375. 1377. 1379. 1381. 1383. 1385. 1387. 1389. 1391. 1393. 1395. 1397. 1399. 1401. 1403. 1405. 1407. 1409. 1411. 1413. 1415. 1417. 1419. 1421. 1423. 1425. 1427. 1429. 1431. 1433. 1435. 1437. 1439. 1441. 1443. 1445. 1447. 1449. 1451. 1453. 1455. 1457. 1459. 1461. 1463. 1465. 1467. 1469. 1471. 1473. 1475. 1477. 1479. 1481. 1483. 1485. 1487. 1489. 1491. 1493. 1495. 1497. 1499. 1501. 1503. 1505. 1507. 1509. 1511. 1513. 1515. 1517. 1519. 1521. 1523. 1525. 1527. 1529. 1531. 1533. 1535. 1537. 1539. 1541. 1543. 1545. 1547. 1549. 15

11/11/11

3. مجلس القضاء الاستئناف المدني الجزائي الادبي العلمي السياسي الاجتماعي البيئي الطبي الرياضي الفني الزراعي الصناعي التجاري البنكي الاسكاني السياحي الثقافي الديني الاجتماعي البيئي الطبي الرياضي الفني الزراعي الصناعي التجاري البنكي الاسكاني السياحي الثقافي الديني

month = ["Jan", "Feb", "Mar", "Apr", "May", "Jun", "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"]

[illegible]

months = [Jan, Feb, Mar, Apr, May, Jun, Jul, Aug, Sep, Oct, Nov, Dec]

$$m = \text{input}(\text{"Month number: "})$$
$$(m) + (n) = m$$

```
m = int(m)
print("Month: ", m, " is ", month[m-1])
```


٨ LF = ["abs", "rand", "pow", "sorted"]
 ٩ LFcopy = LF
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 ١١ LFcopy = LF

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Wednesday
Mercredi

June - Juin

2 DhulQadah 11/1443

Day: 152 / 213

2022

Week 22, Month 6

1443/11 نو القعدة ٢

حزيران - يونيو

الأربعاء

JUN
6

8 range() ← slices
for i in range(9, 11):
 range(s, e, d) ← slice
 s, start, s+2d, s+3d, ...
10
11 list = list(range(0, 25, 2))
 print(list) → [0, 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24]
12

for i in range(0, 11):
 print(base2) → [1, 2, 4, 8, 16, 32, 64, 128, 256, 512, 1024]
 slices ← list of slices
 seq = [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
 slice = seq[0:4]
4 print(seq[0:4]) → [0, 1, 2, 3]
5 print(seq[4:7]) → [0, 1, 2, 3]
6 print(seq[5:]) → [5, 6, 7, 8, 9, 10]
7 print(seq[0:10:2]) → [0, 2, 4, 6, 8]
8 print(seq[-5]) → [0, 1, 2, 3, 4, 5]

Tuesday
Mardi

2

August - Aout

4 Muharram 1 / 1444

Day: 214 / 151

2022

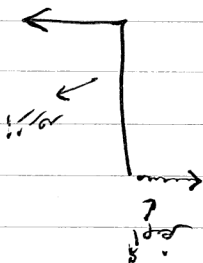
Week 31, Month 8

1444 / 1 / 3

آب - آب

آب - آب

turtle3-3: turtle3-2: the following program
has one turtle, "jared", draw a capital L
in blue and then another, "tina", draw a like
to the west in orange as shown to the left.



```
import turtle  
wn = turtle.Screen()  
jared = turtle.Turtle()  
jared.pensize(10)  
jared.color("blue")  
jared.right(90)  
jared.forward(150)  
jared.left(90)  
jared.forward(75)  
tina = turtle.Turtle()  
tina.pensize(10)  
tina.color("orange")  
tina.left(180)  
tina.forward(75)  
wn.exitonclick()
```


8 (ملاحظة)
 في المثلثات، ملاحظة هامة أخرى أن كل عدد في المثلث هو مجموع العددين اللذين فوقه وبيمينه.
 (Celine, column)

9
 ملاحظة هامة أخرى: كل عدد في المثلث هو مجموع العددين اللذين فوقه وبيمينه.
 ملاحظة هامة أخرى: كل عدد في المثلث هو مجموع العددين اللذين فوقه وبيمينه.
 ملاحظة هامة أخرى: كل عدد في المثلث هو مجموع العددين اللذين فوقه وبيمينه.

12
$$C_n^k = \frac{n!}{k!(n-k)!}$$

$$C_n^k = C_n^{n-k} + C_{n-1}^{k-1}$$

 ← ملاحظة هامة أخرى: كل عدد في المثلث هو مجموع العددين اللذين فوقه وبيمينه.
 ملاحظة هامة أخرى: كل عدد في المثلث هو مجموع العددين اللذين فوقه وبيمينه.

2
$$(1+x)^0 = 1$$

$$(1+x)^1 = 1 + 1 \cdot x$$

$$(1+x)^2 = 1 + 2 \cdot x + 1 \cdot x^2$$

$$(1+x)^3 = 1 + 3 \cdot x + 3 \cdot x^2 + 1 \cdot x^3$$

$$(1+x)^4 = 1 + 4 \cdot x + 6 \cdot x^2 + 4 \cdot x^3 + 1 \cdot x^4$$

$$(1+x)^5 = 1 + 5 \cdot x + 10 \cdot x^2 + 10 \cdot x^3 + 5 \cdot x^4 + 1 \cdot x^5$$

$$(1+x)^6 = 1 + 6 \cdot x + 15 \cdot x^2 + 20 \cdot x^3 + 15 \cdot x^4 + 6 \cdot x^5 + 1 \cdot x^6$$

| | | | | | | | |
|---|---|---|----|----|----|---|---|
| 8 | 1 | 6 | 15 | 20 | 15 | 6 | 1 |
| 7 | 1 | 5 | 10 | 10 | 5 | 1 | |
| | 1 | 4 | 6 | 4 | 1 | | |
| | 1 | 3 | 3 | 1 | | | |
| | 1 | 2 | 1 | | | | |
| | 1 | 1 | | | | | |
| | 1 | | | | | | |
| | 1 | | | | | | |

٣١
الاثنين

أيار - مايو

1 ذو القعدة 1443/11

Week 22, Month 5

2022

Day: 151 / 214

1 DhulQadah 11/1443

May - Mai

Tuesday 31
Mardi

5
MAY

٨
T. هذا اليوم كرمنا بـ...
٥
n x n
٤
n = 4
٣
...
٢
...
١
...

11 $T_{i-1} = T_{i-1} + T_{i-1}$

$n=13$ # ...

...

$T = [0 \text{ for } i \text{ in range}(0,n)]$ for j in range($0,n$)

...

for i in range($0,n$):

for j in range($0,n$):

$T[i][j] = T[i-1][j] + T[i-1][j-1]$

...

for i in range($0,n$):

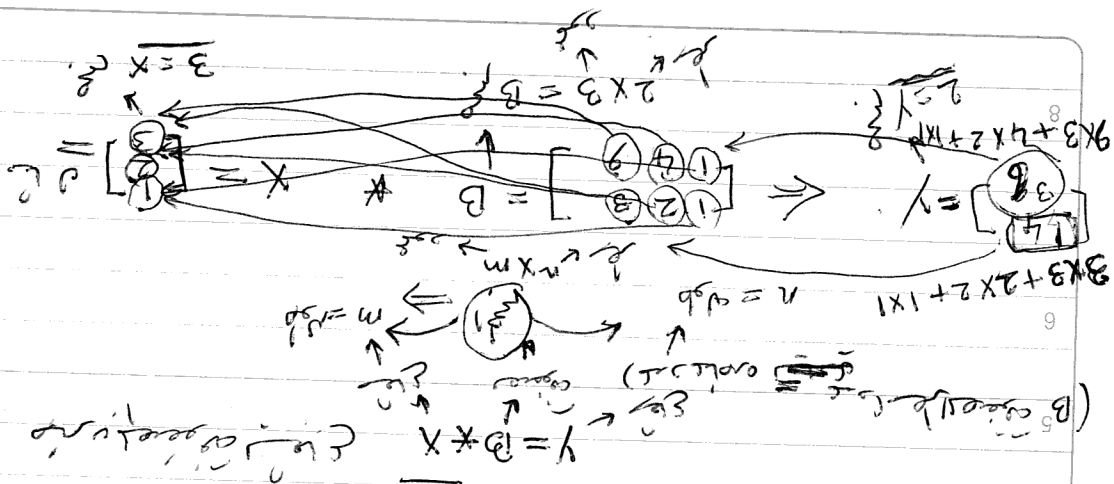
row = #

for i in range($0,n$):

row = #

$$\text{row} + = n + 81r(T[4][5])$$

print(row)

[illegible]

B = [1, 2, 3]
[1, 4, 9]
X = [1, 2, 3]

8

7

6

5

4

3

2

1

12

11

10

9

8

```

B = [1, 2, 3]
[1, 4, 9]
X = [1, 2, 3]

n = len(B)  => # n = 2
m = len(X)  => # m = 3

Y = [0, 0] # m * 2
for i in range(n):
    for j in range(m):
        Y[i] += B[i] * X[j]

print(m * B)
for i in range(0, len(B)):
    print(B[i])
print("X")
print("Y")
print("\nY = B * X")
    
```

{w} => B

[1, 2, 3]
[1, 4, 9]

X

May - Ma!

**Saturday
Samedi!**

87



2. $\int_0^1 x^2 dx = \frac{1}{3}$

۱۳۳۱/۱۰ ۱۲۱۱ ۲۷

Week22, Months

2202

Day: 148 / 217

27 Shawal 10 / 1443

$$Y = B \cdot X$$
[illegible]

$N = 16$

$N=10$
 $E_{\text{env}} = [! \text{ for } i \text{ in range}(0, N+1, 2)]$
 $\Rightarrow E_{\text{env}} = [0, 2, 4, 6, 8, 10, 12, 14, 16]$

$$\Rightarrow E_{\text{AVA}} = [0, 2, 4, 6, 8, 10, 12, 14, 16]$$
[illegible]
$$T = T$$

[0 for i in range(0,n)]

for j in range(0, n)]

3. [A] [B] [C] [D] [E] [F] [G] [H] [I] [J] [K] [L] [M] [N] [O] [P] [Q] [R] [S] [T] [U] [V] [W] [X] [Y] [Z]

$$T = T$$

for i in range(0, n):
for j in range(0, n):

for n in range(0, n)

$[1, 0, 0], [1, 0], [2, 0], [3, 0], [4, 0]$
 $[1, 0, 1], [1, 1], [2, 1], [3, 1], [4, 1]$
 $[1, 0, 2], [1, 2], [2, 2], [3, 2], [4, 2]$
 $[1, 0, 3], [1, 3], [2, 3], [3, 3], [4, 3]$
 $[1, 0, 4], [1, 4], [2, 4], [3, 4], [4, 4]$

12436, Sami Ahmad, 80.8, 90.5, 83.7, Success
12440, Anas Mansour, 50.4, 80.5, 59.4, Fail
13822, Moustafa Zein, 39.0, 95, 55.8, Fail

12436, Sami Ahmad, 80.8, 90.5
12440, Anas Mansour, 50.4, 80.5
13822, Moustafa Zein, 39.0, 95, 55.8

12436, Sami Ahmad, 80.8, 90.5
12440, Anas Mansour, 50.4, 80.5
13822, Moustafa Zein, 39.0, 95, 55.8

String, float, int, ...

5 MAY

الطلاب

40 < 40 / 2

40 < 40 / 2

60 < 60 / 3

id , name Xam , Pract

Students = [[12436, "Sami Ahmed", 80.8, 90.5], [12440, "Anas Mansour", 50.4, 80.5], [13822, "Moustafa Zein", 57.2, 70], [12521, "Ahmad Alos", 39.4, 80.5], [16621, "Zouhair Dahrouj", 55.2, 35]]

for i in range(0, len(students)):

Xam = students[i][2]

Pract = students[i][3]

Score = Xam * 0.70 + Pract * 0.30

Success = (Xam > 40) and (Pract > 40) and (Score > 60)
Print students[i], "%0.1f" % Score, "%0.1f" % Success
if Success else "Fail"

[12436, "Sami Ahmed", 80.8, 90.5], 83.7, Success

[12440, "Anas Mansour", 50.4, 80.5], 59.4, Fail

[13822, "Moustafa Zein", 57.2, 70], 61.0, Success

[12521, "Ahmad Alos", 39.4, 80.5], 51.7, Fail

[16621, "Zouhair Dahrouj", 55.2, 35], 49.1, Fail

٨

Bubble sort

٩

string = ["Zouhair", "Ahmed", "Zaid", "Omar"]
 print(string)
 for i in range(0, len(string)):
 for j in range(i+1, len(string)):
 if (string[i] > string[j]):
 t = string[i]
 string[i] = string[j]
 string[j] = t
 print("Sorted by bubble sort")
 print(string)

٣

=> ["Zouhair", "Ahmed", "Zaid", "Omar"]
 Sorted by bubble sort:
 ["Ahmed", "Mari", "Omar", "Zaid", "Zouhair"]

٤

Sorted array: ["Zaid", "Omar", "Ahmed", "Mari", "Zouhair"]
 Sorted (string)

MAY 5

May - Ma!

Tuesday
24
Mardi!

1. Mathematical

٣٢

ᠭᠠᠨ - ᠨᠠᠭ

2022

Day: 144 / 221

Week21, Months

۱۳۳۵/۱۰ ۱۲۳۱

1. Qualitätsmanagement
 2. Produktionsmanagement
 3. Personalmanagement
 4. Finanzmanagement
 5. Marketingmanagement
 6. Strategisches Management
 7. Operatives Management
 8. Strategisches Marketing
 9. Operatives Marketing
 10. Strategisches Personalmanagement
 11. Operatives Personalmanagement
 12. Strategisches Finanzmanagement
 13. Operatives Finanzmanagement
 14. Strategisches Produktionsmanagement
 15. Operatives Produktionsmanagement

① strings are stored in memory as array of characters

String 1 String 2

newString = "This is a string literal"

literal one = "1111 MySylhem 11 My Directory 11 Prog.cs"

Verbal interaction = 11 Phys system directly

Literal Two = "Line one in Line Two"

1. $\frac{1}{x^2} = x^{-2}$

* $\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}} \right) = \frac{\partial L}{\partial x}$

1. $\frac{1}{m} \left(\frac{1}{m} \right)$

ستة عشر إلى ثمانية عشر

3 জীববিগ্নি মুদ্রণ কেন্দ্র

Star [1] 2x

11575 11576 11577 11578 11579 11580 11581 11582 11583 11584 11585 11586 11587 11588 11589 11590 11591 11592 11593 11594 11595 11596 11597 11598 11599 11600 11601 11602 11603 11604 11605 11606 11607 11608 11609 11610 11611 11612 11613 11614 11615 11616 11617 11618 11619 11620 11621 11622 11623 11624 11625 11626 11627 11628 11629 11630 11631 11632 11633 11634 11635 11636 11637 11638 11639 11640 11641 11642 11643 11644 11645 11646 11647 11648 11649 11650 11651 11652 11653 11654 11655 11656 11657 11658 11659 11660 11661 11662 11663 11664 11665 11666 11667 11668 11669 11670 11671 11672 11673 11674 11675 11676 11677 11678 11679 11680 11681 11682 11683 11684 11685 11686 11687 11688 11689 11690 11691 11692 11693 11694 11695 11696 11697 11698 11699 11700 11701 11702 11703 11704 11705 11706 11707 11708 11709 11710 11711 11712 11713 11714 11715 11716 11717 11718 11719 11720 11721 11722 11723 11724 11725 11726 11727 11728 11729 11730 11731 11732 11733 11734 11735 11736 11737 11738 11739 11740 11741 11742 11743 11744 11745 11746 11747 11748 11749 11750 11751 11752 11753 11754 11755 11756 11757 11758 11759 11760 11761 11762 11763 11764 11765 11766 11767 11768 11769 11770 11771 11772 11773 11774 11775 11776 11777 11778 11779 11780 11781 11782 11783 11784 11785 11786 11787 11788 11789 11790 11791 11792 11793 11794 11795 11796 11797 11798 11799 11800 11801 11802 11803 11804 11805 11806 11807 11808 11809 11810 11811 11812 11813 11814 11815 11816 11817 11818 11819 11820 11821 11822 11823 11824 11825 11826 11827 11828 11829 11830 11831 11832 11833 11834 11835 11836 11837 11838 11839 11840 11841 11842 11843 11844 11845 11846 11847 11848 11849 11850 11851 11852 11853 11854 11855 11856 11857 11858 11859 11860 11861 11862 11863 11864 11865 11866 11867 11868 11869 11870 11871 11872 11873 11874 11875 11876 11877 11878 11879 11880 11881 11882 11883 11884 11885 11886 11887 11888 11889 11890 11891 11892 11893 11894 11895 11896 11897 11898 11899 11900 11901 11902 11903 11904 11905 11906 11907 11908 11909 11910 11911 11912 11913 11914 11915 11916 11917 11918 11919 11920 11921 11922 11923 11924 11925 11926 11927 11928 11929 11930 11931 11932 11933 11934 11935 11936 11937 11938 11939 11940 11941 11942 11943 11944 11945 11946 11947 11948 11949 11950 11951 11952 11953 11954 11955 11956 11957 11958 11959 11960 11961 11962 11963 11964 11965 11966 11967 11968 11969 11970 11971 11972 11973 11974 11975 11976 11977 11978 11979 11980 11981 11982 11983 11984 11985 11986 11987 11988 11989 11990 11991 11992 11993 11994 11995 11996 11997 11998 11999 12000 12001 12002 12003 12004 12005 12006 12007 12008 12009 12010 12011 12012 12013 12014 12015 12016 12017 12018 12019 12020 12021 12022 12023 12024 12025 12026 12027 12028 12029 12030 12031 12032 12033 12034 12035 12036 12037 12038 12039 12040 12041 12042 12043 12044 12045 12046 12047 12048 12049 12050 12051 12052 12053 12054 12055 12056 12057 12058 12059 12060 12061 12062 12063 12064 12065 12066 12067 12068 12069 12070 12071 12072 12073 12074 12075 12076 12077 12078 12079 12080 12081 12082 12083 12084 12085 12086 12087 12088 12089 12090 12091 12092 12093 12094 12095 12096 12097 12098 12099 12100 12101 12102 12103 12104 12105 12106 12107 12108 12109 12110 12111 12112 12113 12114 12115 12116 12117 12118 12119 12120 12121 12122 12123 12124 12125 12126 12127 12128 12129 12130 12131 12132 12133 12134 12135 12136 12137 12138 12139 12140 12141 12142 12143 12144 12145 12146 12147 12148 12149 12150 12151 12152 12153 12154 12155 12156 12157 12158 12159 12160 12161 12162 12163 12164 12165 12166 12167 12168 12169 12170 12171 12172 12173 12174 12175 12176 12177 12178 12179 12180 12181 12182 12183 12184 12185 12186 12187 12188 12189 12190 12191 12192 12193 12194 12195 12196 12197 12198 12199 12200 12201 12202 12203 12204 12205 12206 12207 12208 12209 12210 12211 12212 12213 12214 12215 12216 12217 12218 12219 12220 12221 12222 12223 12224 12225 12226 12227 12228 12229 12230 12231 12232 12233 12234 12235 12236 12237 12238 12239 12240 12241 12242 12243 12244 12245 12246 12247 12248 12249 12250 12251 12252 12253 12254 12255 12256 12

[illegible]

(6) $\frac{d}{dx} \left(\frac{x^2 + 1}{x - 1} \right)$

مجلس الشورى

```
newString = "Jon has " + str(34) + " books."
```

$$b = \text{new mas} + \text{str}(34) + \text{"books"}$$

newString = "Don has 33 books" format(100)

newstring2 = "Itan has 3X3 blocks" format(3.4)

new string 3 = "Jon has 30 books and 1/2 pen" form H[3]

```
newString3 = "Jon has 203books and 513 pen".format(34, 17, 5)
```

X = 123
print("The value is %d" % X) →
print("The value is %06d" % X)

X = 123.4567
print("The value is %f" % X)
print("The value is %12.6f" % X)

The value is 123
The value is 000123
The value is 123.456700
The value is 123.457

print("The value is %d" % X)
newstrings = "John has %d books and %d pen" % (34, 45)

X = 123
print("The value is %d" % X)
print("The value is %06d" % X)

X = 123.4567
print("The value is %12.6f" % X)
print("The value is %12.3f" % X)

Sunday
Dimanche

May - Mai

21 Shawal 10 / 1443

Day: 142 / 223

2022

Week 21, Months

21 شوال 10 / 1443

أيار - مايو

٢٢
الأيار

| | | |
|----|-------------------------|--|
| 8 | The value is 123 | |
| 9 | The value is 123,456700 | |
| 10 | The value is 000123 | |
| 11 | The value is 123,457 | |
| 12 | The value is 123,457 | |
| 1 | The value is 123,457 | |
| 2 | The value is 123,457 | |
| 3 | The value is 123,457 | |
| 4 | The value is 123,457 | |
| 5 | The value is 123,457 | |
| 6 | The value is 123,457 | |
| 7 | The value is 123,457 | |
| 8 | The value is 123,457 | |

Sunday
Dimanche

6

٦ الأحد

February - Fevrier

شباط - فبراير

5 Rajab 7 /1443

Day: 37 / 328

2022

Week6, Month2

٥ رجب ١٤٤٣

8

9

10

11

12

1

2

3

4

5

6

7

8

Saturday
Samedi

5

February - Fevrier

2
FEB

4 Rajab 7 /1443

Day: 36 / 329

2022

Week6, Month2

شباط - فبراير

٤ رجب ١٤٤٣/٧

السبت ٥

8

9

10

11

12

1

2

3

4

5

6

7

8

```
elif playerMove == 'r' and  
    computerMove == 'S':
```

```
    Print('you win!')
```

```
    wins = wins + 1
```

```
elif playerMove == 'p' and
```

```
    computerMove == 'r':
```

```
    print('you win!')
```

```
    wins = wins + 1
```

```
elif playerMove == 'r' and
```

```
    computerMove == 'P':
```

```
    print('you lose')
```

```
    losses = losses + 1
```

```
elif playerMove == 'p' and
```

```
    computerMove == 'S':
```

```
    Print('you lose')
```

```
    losses = losses + 1
```

```
elif playerMove == 'S' and
```

```
    computerMove == 'r':
```

```
    Print('you lose')
```

```
    losses = losses + 1
```

#Display what the player

8 chose:

if playerMove == 'r':

9 print('Rock versus...')

elif playerMove == 'p':

10 print('PAPER versus...')

elif playerMove == 's':

11 print('SCISSORS versus...')

12 #Display what the computer chose:

randomNumber = random.randint(1, 3)

1 if randomNumber == 1:

computerMove = 'r'

2 print('Rock')

elif randomNumber == 2:

3 computerMove = 'p'

print('PAPER')

4 elif randomNumber == 3:

computerMove = 's'

5 print('SCISSORS')

#Display the record the win/loss/tie:

if PlayerMove == computerMove:

7 print('It is a tie!')

ties = ties + 1

8

الأكور

import random, sys

print('Rock, PAPER, SCISSORS')

#These variables keep track
of the number of wins,
losses, and ties.

wins = 0

losses = 0

ties = 0

(while True: #The main
game loop.

print('%s wins, %s Losses,

%s Ties' % (wins,

losses, ties))

while True: #The player
input

print('Enter your move:

(r)ock (p)layer

(s)cissors or (q)uit)

playMove = input()

if playMove == 'q':

sys.exit()

if playMove == 'r' or

playMove == 'P' or

playMove == 's':

Break

print('Type one of
r, p, s, or q')

Thursday
Jeudi

3

February - Fevrier

Ties

الخميس ٣

شباط - فبراير

2
FEB

2 Rajab 7/1443

Day: 34 / 331

2022

Week5, Month2

٢ رجب ١٤٤٣

else:

8 print('Nope. The number
I was thinking
9 of was' + str(
secretNumber))

10

ربنا لا اله الا انت سبحانك اني كنت من الظالمين (البقرة)

11

يكون الزرع تالفاً

Rock, PAPER, SCISSORS

0 wins, 0 Losses, 0 Ties

Enter your move: (r)ock (p)aper
(s)ciissors or (q)uit

p

PAPER
SCISSORS versus...

PAPER

It is a tie!

0 wins, 1 Losses, 1 Ties

Enter your move: (r)ock (p)aper
(s)ciissors or (q)uit

s

SISSORS versus...

PAPER

You win!

1 wins, 1 Loss, 1 Ties

Enter your move: (r)ock (p)aper
(s)ciissors or (q)uit

q

This is a guess the
number game.

import random

secretNumber = random.

randint(1, 20)

Print('I am thinking of
a number between
1 and 20.')

Ask the Player to guess
6 times.

for guessesTaken in
range(1, 7):

print('take a guess')

guess = int(input())

if guess < secretNumber

print('Your guess
is too low')

elif guess > secretNumber

print('Your guess
is too high')

else:

break # this is
the correct

if guess == secretNumber

print('Good job!

you guessed my number

in' + str(guessesTaken)

Wednesday
Mercredi

2

February - Fevrier

guess

الأربعاء ٢

شباط - فبراير

1 Rajab 7/1443

Day: 33 / 332

2022

Week5, Month2

1 رجب ١٤٤٣/٧

8

برنامجي قصير: تخمين الرقم.

أنا أفكر في عدد بين 1 و 20.

9 I am thinking of a number
between 1 and 20.

10 Take a guess

10

11 your guess is too low.

Take a guess

12 15

your guess is too low

1 Take a guess

17

2 your guess is too high

Take a guess

3 16

Good job! you guessed my

4 number in 4 guesses.

5

6

dition

guess.

Der?

you

* مثال آخر على استخدام مودول random

import random, sys, os, math

من random لاستيراد

* مودول random

from random import *

ونصل لعدد 1 كحد أقصى

استخدام

import random

* ايجاد البرنامج بكبر الآلة

sys.exit()

- كما في استخدام sys

بعد استيراد

من

import sys

while true:

print('Type exit to exit.')

response = input()

if response == 'exit':

sys.exit()

print('you typed' +

response + '.')

| * استيراد لبرمجيات (المودولات) | مثال آخر |
|---|---|
| <p>8
import random
9
for i in range(5):
 Print(random.randint(1,10))</p> | <p>For i in range(0, 10, 2)
 Print(i)</p> <p>↓</p> <p>0
2
4
6
8</p> |
| <p>10
البرمجة عند نقطة استيراد لبرمجيات لا يكون لها
 كمبيوتر مودولون لبرمجيات الحقيقة
os.py و sys.py و random.py
12
أو math.py
وإذا قمنا بكتابة برنامجه
random.py فسنرى رة في رة في برنامجه آخر
بأننا Import random عند نقطة
هذا البرنامج يستورد برنامجه random.py
بواسطة العبارة random لبرمجيات
رنا برنامجه فها
3
AttributeError: module 'random'
4
has no attribute 'randint'
هذه الرسالة كما في رسالة random
* كذا نستخدم اسماء في دوال البرمجة
6
input() أو print() لبرمجيات
وغيرها
7</p> | <p>يمكننا أن نستخدم عدد سالب
في مكان عدد فخران في range()
فعل فلة بعد كل ما نري به
من الصغرى
for i in range(5, -1, -1)
 print(i)</p> <p>↓</p> <p>5
4
3
2
1
0</p> |
| <p>8
4
1
8
4
1</p> | <p>عند تنفيذ هذا البرنامج سنحصل على النتيجة
↓</p> |

2022

| | | |
|----|---|---|
| 8 | while
كانت أيضاً استخدمت while
لكتابة البرنامج ليقرأ اسمي ويكتب لي
Jimmy خمسة مرات
(كانت while for نسخة أخرى) | break
استخدمت أيضاً break
For و continue
* continue
التي تأتي بعد while
for |
| 10 | Print('My name is')
i=0 | * تأتي في نسخة لبرنامجي
break و continue |
| 11 | while i < 5:
print('Jimmy Five Times') | ضمن while
for و إذا طرأ |
| 12 | + str(i) + ' ')
i = i + 1 | استخدمت مكان آخر
سيعطين رسالة خطأ البايثون |
| 1 | و عند تنفيذ البرنامج سيكون ينتج مثل هذه for
(في البداية فاقبول عبد) | ربما لي مجموع لرقم مع صفها
من 100 < 10 |
| 3 | البداية والنهاية والخطوة في range()
range() | ① total = 0
② for num in range(101):
③ total = total + num
④ print(total) |
| 4 | for i in range(12, 16):
print(i) | يجب أن يكون ينتج 5,050
* توجد طريقة أخرى لحساب
وهي
هكذا،
1 + 100, 2 + 99, 3 + 98
... 50 + 51
و ما أن 50 x 101
5050 و المجموع كل الأرقام
من 100 < 10 |
| 5 | ↓ | |
| 6 | 12 | |
| | 13 | |
| 7 | 14 | |
| | 15 | |
| 8 | | |

name اذا دخلت اسمك في حقله
 ① True = while حاليه
 ② numofGuests نسبه 0
 ③ True
 * عليك ان ادخل اسمك في حقله
 numofGuests != 0 و not name
 True
 False
 12

طابق For و range
 for i in range(5):
 ①
 ②
 ③
 ④
 ⑤
 ⑥
 ⑦
 ⑧
 ⑨
 ⑩
 ⑪
 ⑫
 ⑬
 ⑭
 ⑮
 ⑯
 ⑰
 ⑱
 ⑲
 ⑳
 ㉑
 ㉒
 ㉓
 ㉔
 ㉕
 ㉖
 ㉗
 ㉘
 ㉙
 ㉚
 ㉛
 ㉜
 ㉝
 ㉞
 ㉟
 ㊱
 ㊲
 ㊳
 ㊴
 ㊵
 ㊶
 ㊷
 ㊸
 ㊹
 ㊺
 ㊻
 ㊼
 ㊽
 ㊾
 ㊿

print('My name is')
 for i in range(5):
 print('Jimmy five times
 (' + str(i) + '))

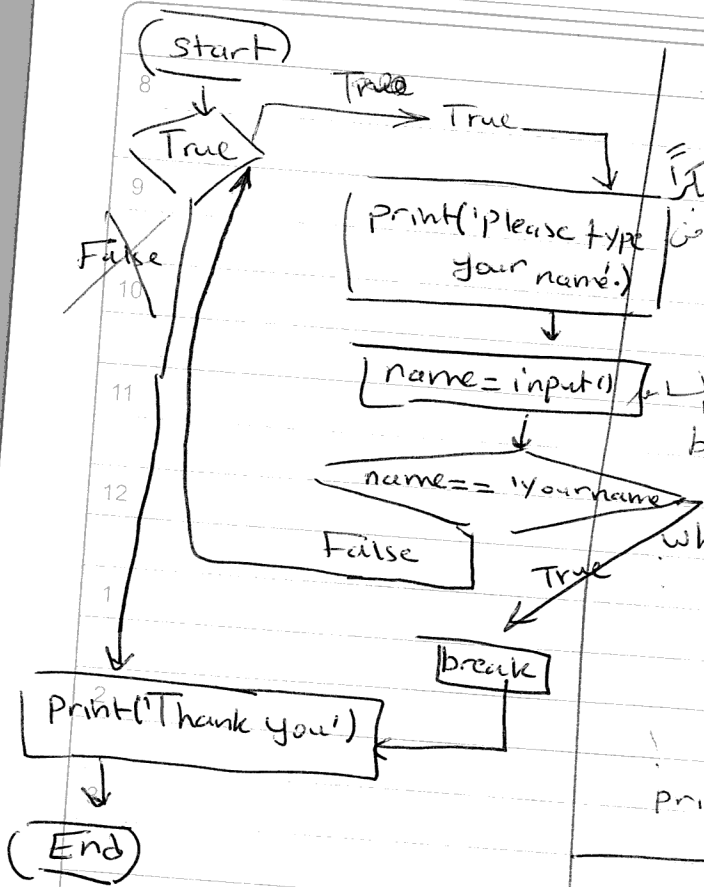
My name is
 Jimmy five times (0)
 (1)
 (2)
 (3)
 (4)

False و True

القيم 0 او 0.0 او 1
 او حقله فارغ
 False
 True
 صان

name = ''
 ① while not name:
 print('Enter your name')
 name = input()
 print('How many
 guests will you
 have?')
 صان

numofGuests = int(input())
 ② if numofGuests:
 ③ print('Be sure
 to have
 enough room
 for all your
 guests.')
 print('Done')



عبارت break

وهذا الخروج من الحلقة while مبكراً
حيث عند هذا وقتاً يخرج البرنامج من
الحلقة فوراً.

في المثالي نفضل كلاً من البرنامج
بأنه سينتج العبارة break

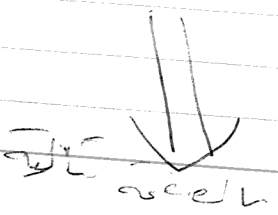
```

while True
    print('please type your name')
    name = input()
    if name == 'your name':
        break
    print('Thank you').
  
```

البيان Continue

تستخدم عند break داخل الحلقات
معناها عند تنفيذ البرنامج إليها فإنه
تقفز مباشرة لبيان اللاحقة دون
تنفيذ سطر اللاحقة.

مثال البرنامج التالي swordfish.py



ملحوظة: إذا ملق البرنامج بصورة
ملف في كذا مرة أكتب
Ctrl + C من أجل
الإنهاء.

```

while True:
    print('Hello, world!')
  
```

Sunday
Dimanche

30

January - Janvier

27 Jumada-2 6/1443

Day: 30 / 335

2022

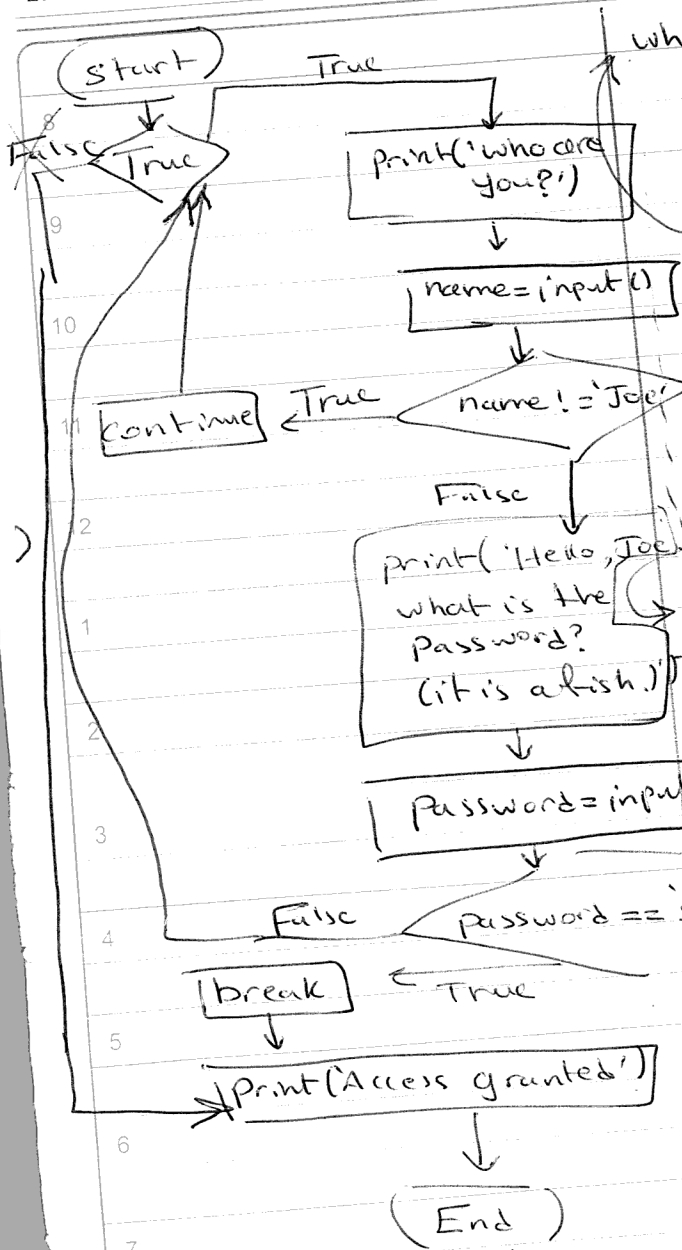
Week5, Month1

٢٧ جمادى الثاني ١٤٤٣/٦

besides

الإحد ٣٠

كانون الثاني - يناير



```
while True:
    print('who are you?')
    name = input()
    if name != 'Joe':
        continue
    print('Hello, Joe. what is
    the password? (It's
    a fish.)')
    password = input()
    if password == 'swordfish':
        break
```

```
print('Access granted')
```

- who are you?
I'm fine, thanks. who are
you?
- who are you?

Joe
- Hello, Joe. what is
the password? (It's
a fish.)

Mary

- who are you?

Joe

- Hello Joe. what is
the password? (It's
a fish.)

Swordfish

- Access Granted.

elif $\frac{1}{2} \leq \frac{a}{b} < 1$ then

عن أبي عبد الله (عليه السلام) قال: «ما

وہ کرانے کے لیے

تجارتی اور نو مالیاتی حلال فی حال

oer Kij is een, True

November 21st 1902

1. चर

name = 'Carol'

age = 3000

if name == 'Alice':

```
print('Hi, Alice')
```

elif age < 12:

```
Print('You are not Alice,  
kiddo')
```

Kiddo')

⑤ $\text{einf. age } > 100:$

```
print('You are not-Alice  
grannie,')
```

elif age > 2000:

```
print('Unlike you, Alice  
is not an Undead,  
immortal vampire')
```

immortal vampire.)

7 else:

```
print('you are neither  
Alice nor a little kid.')
```

Alice, kiddo!

7 else:

```
print('you are neither
```

8 Alice not a little kid.')

| | |
|---|---|
| <p>8 while loop في لغة بايثون</p> <p>name = ''</p> <p>9 while name != 'your name':</p> <p>10 print('please type your name.')</p> <p>name = input()</p> <p>11 print('Thank you!')</p> | <p>while loop في لغة بايثون</p> <p>1 if name == 'your name':</p> <p>spam = 0</p> <p>if spam < 5:</p> <p>print('Hello, world')</p> <p>spam = spam + 1</p> |
| <p>12</p> <p>(start)</p> <p>1 name != 'your name'</p> <p>2 False</p> <p>3 print('please type your name.')</p> <p>4 name = input()</p> <p>5</p> <p>6 print('Thank you!')</p> <p>7</p> <p>8</p> | <p>2) while loop في لغة بايثون</p> <p>spam = 0</p> <p>while spam < 5:</p> <p>print('Hello, world')</p> <p>spam = spam + 1</p> <p>تسمى البراعة في</p> <p>في لغة بايثون</p> <p>Hello, world</p> <p>5</p> |

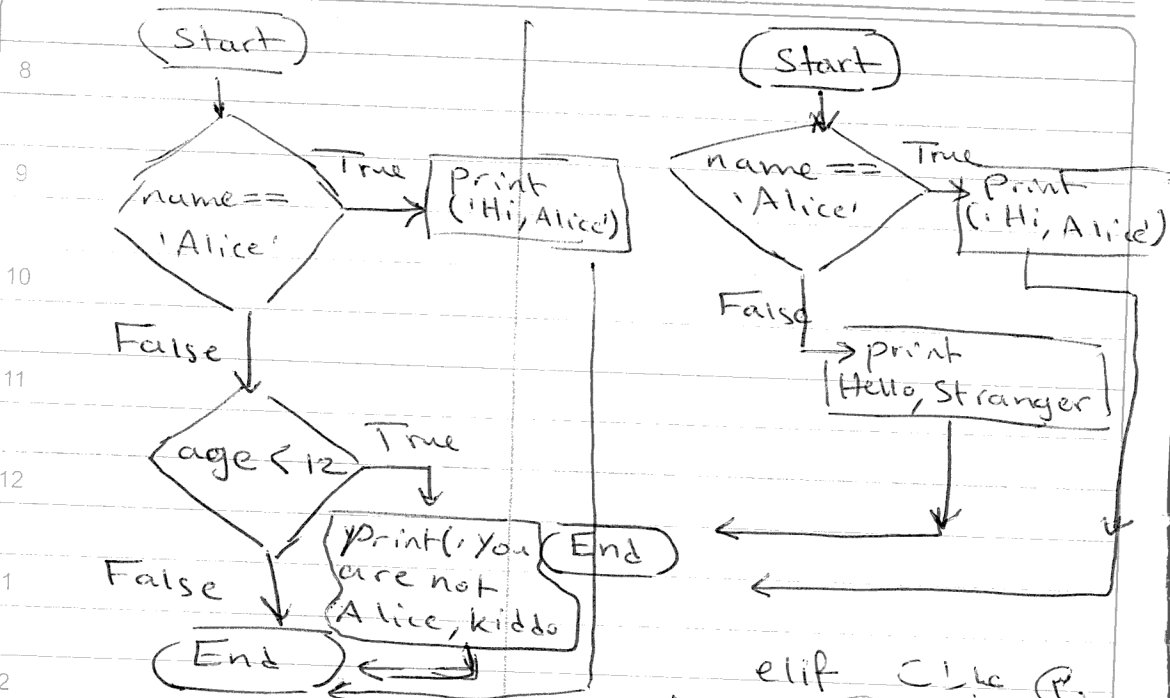
ابن immortal

vampire vampire
grannie جدة

Kiddo

الثلاثاء ٢٥

كانون الثاني - يناير

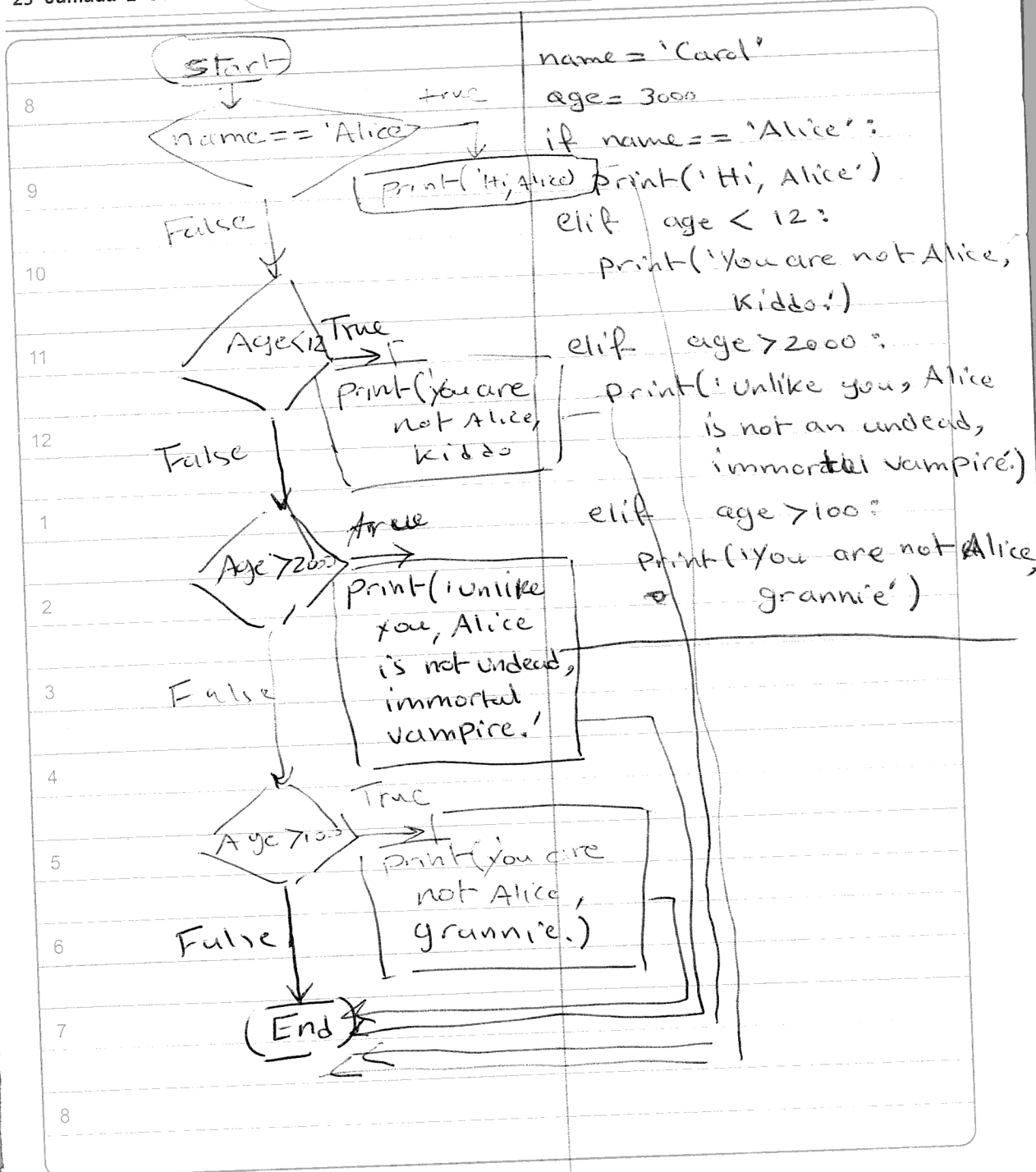


elif elif
else if

Guaranteed

elif
True elif
elif

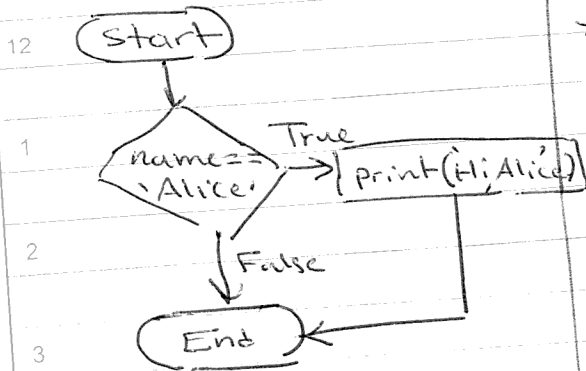
if name == 'Alice':
print('Hi, Alice')
elif age < 12:
print('You are not Alice,
kiddo.')



٨ if name == 'Alice' (1)

الزلافة
في لوحة بيضاء

١٠ if name == 'Alice' (1)
Print('Hi, Alice.')



٤ else (2)
٥ if name == 'Alice':
Print('Hi, Alice')

٦ else:
Print('Hello, Stranger')

٨

True or True → True (or)
True or False → True
False or True → True
False or False → False

(not)
not True → False
not False → True
>>> (4 < 5) and (5 < 6)
True
>>> (4 < 5) and (9 < 6)
False
>>> (1 == 2) or (2 == 2)
True

>>> 2+2 == 4 and not 2+2 == 5
and 2+2 == 2+2

True
في لوحة البيانية لكانون الثاني
في لوحة بيضاء

not (1)
and (2)
or (3)


```
>>> spam = int('spam')
```

```
8 >>> spam
```

```
101
```

دعنا الآن نكتب مع القيمة المخزنة
داخل spam (نكتب spam) لندرج
بالتالي

```
>>> spam + 10/5
```

```
11 202.0
```

لاحظ إذا قمنا بكتابة

دالة لنكتب تحويل لندرج

```
1 <= int('99.99')
```

```
2
```

خطأ

ValueError: invalid literal
for int() with base 10:

```
1 99.99'
```

```
4
```

```
int('twelve')
```

```
5
```

خطأ

نفس الدالة int() أيضاً إذا كنا نكتب

عدد بـ الرقم ذو اعشار لندرج
الرقم أخفض

```
int(7.7) => 7
```

```
8 int(7.7) + 1 => 8
```

```
>>> str(29)
```

```
1 291
```

```
>>> print('I am' + str(29) +  
' years old.')
```

```
I am 29 years old.
```

← str() تحويل لندرج

← int() لندرج

← float() تحويل لندرج

```
str(0) => '0'
```

```
str(-3.14) => '-3.14'
```

```
int('42') => 42
```

```
int('-99') => -99
```

```
int(1.25) => 1
```

```
int(1.99) => 1
```

```
float('3.14') => 3.14
```

```
float(10) => 10.0
```

نفس الدالة input() دائماً

قيمة نصية، من لندرج

نما

```
>>> spam = input()
```

```
101
```

```
>>> spam
```

```
101
```

لاحظ

8 errand كحة

diamonds ماسة - الماس

9 intentionally عمدًا - قصداً

10 >>> true

→ خطأ

11 NameError: name 'true' is not defined.

12

>>> True = 2+2

1

← خطأ

SyntaxError: can't assign

2

to keyword.

3 تبارك وتعالى، على رسلكم

لنعم لعلنا نرى في لقاءك علينا نرى

في عبادات

معادلات مقارنة (المعادلات المتكافئة)

= 1 مساوي

6 != 1 لساوي

< اقل من

7 > اكبر من

<= اقل من او مساوي

8 >= اكبر من او مساوي

print('What is your age?')

myAge = input()

print('You will be ' + str(int(myAge) + 1) + ' in a year.')

نظام فوري (مترجم) ليس مترجم

42 == '42' ⇒ False

42 == 42.0 ⇒ True.

42 == 0042.000 ⇒ True.

* skim تخطي - يهين

round() الدالة

* التكميم لبيانات (المتن)

True الصواب

False

False True

↓
حرف كبير

* >>> span = True

>>> span

True

⑦ print('That password
is one that an
idiot puts on their
luggage.')
else:
⑧ print('Access denied')

every once in a while as
you type them.

* تسمى البرنامج تسمى فورا
F5 - رجع البرنامج لاول الجرافيك
تدوير

Hello, world!

what is your name?

Al

It's good to meet you, Al.

The Length of your name
is :

2

what is your age?

4

You will be 5 in a year.

terminates . نتيجه

* يمكن رؤية تنفيذ البرنامج على
python Tutor visualization Tool
على

<http://pythontutor.com>

ويمكن رؤية تنفيذ برنامج فاصد

<http://author.com/hellopy/>

نص على انهم ليسوا بملفات بل بترقية
ويمكن رؤية انهم ليسوا بملفات بل بترقية
البرنامج

الملاحظات Comments

وهذا برنامج # قبل الجرافيك
وهذا هو الجرافيك # قبل الجرافيك

##

hash mark

commenting out code

وهو وضع # قبل الجرافيك
على السطر وتكون الجرافيك
البرنامج

Figure out كيف - نفهم

print()

Function هو والبرنامج
print('Hello, world')

argument نرى 'Hello, world!'

8 * بعد print('Hello') من المربع على
علامة إقفال.
9 * يمكننا أيضاً استعمال دالة print() لرفع
خط فارغ على الشاشة.

10 الدالة input() تستغرقه الدالة! المتغير لتقبل بعض النص
من لوحة المفاتيح وليكتب Enter

12 myName = input()
فيكون محتوى
1 بعد string

3 الدالة len() تعطينا قيمة string
أو قيمة عددية string

4 len('Hello') ⇒ 5
len('') ⇒ 0
5 تسمى الدالة print() بتربطها بإقنية
عدد صيغة أو قيم string.
6 لكن لاحظ الخطأ المتولد عند ادخال سطر
print('I am' + 29 + 'years old')

7 Type error: خطأ
can only concatenate str(not "int")
8 to str.

الخطأ المتولد ليس عند Print() إنما عند التعبير الذي تماد
تربيته print() حيث أنه تماد على تعده الفا
إذا أوقلت التالي!

>>> 'I am' + 29 + 'years old'
← خطأ

TypeError: can only concatenate str(not "int") to str.

أعطى بايثون خطأ لأن المتداد +
ممكن استعمله فقط لجمع عددين
صحيين مثلاً أو دمج سلسلتين
لأنه جمع عددين مع سلسلة
لأن هذا غير مقبوع في
بايثون.

يمكن إصلاح ذلك باستخدام
دالة str() لتحويل العدد الصحيح
كما يلي: str()

الدالة int() و str()
float()

4 إذا رغبت برفع عدد صحيح
من 29 مع سلسلة
تربطها print() ستحتاج
إلى التحويل على النحو
الآتي: str(29) + 'years old'

Explicitly

مکمل واضح

* it makes sense

8 Dissecting

١٢٤

repli'cate تَکْثِیرٌ وَ تَکْرِیرٌ

arbitrary $\frac{d}{dx}$

fractional جزء

assignment - تعین

* متغيرات مختلفة عن كود integer
string، وغيرها كمتغيرات
float(), int(), str()

float(), int(), str()

* عن أبي حمزة الثمالی * (الضرب) به

- integer $\rightarrow$ string

فأما كمال الله عدد مرات ¹²integer

1 >>> 'Alice' * 5

'AliceAliceAliceAliceAlice'

2 وهذا يسمى الاستماع.

عَنْ اسْتِثْنَاءِ الْعَاطِلِ * فَقَطْ ٣

قَمِيمِينَ عَدَسِينَ (لِلْفَصْلِ) أَوْ
قَمِيمَةَ String واحدة وقَمِيمَةَ صَمِيمَةَ

integer واحد (مفرد) استماع
العدد والرقم باليونان
سعد سعد فطاح

6
 >>> 'Alice' * 'Bob'



377 Alice, ¥ 5.0

→

8 integer

Current_balance (2)

Current-balance (X) $\frac{I_{\text{current}}}{I_{\text{balance}}}$

account 4 (✓)

4 account (K) لود کنندہ کے نام

-42 (✓)

42 (X) لا يمكن أن يبدأ قسم

TOTAL - sum (✓)

TOTAL - \$um (x) غير مجمع استقام
المطابق الخاصة شير

hello (✓)

hello, (x) غرض سے اس کا نام
الٹی لکھی جاتی ہے اس لیے

Spam, SPAM, spam, SPAM

Convention عرف اتفاقية

8

* عرف في لايون - هذا اسم لبرمجة

9

* camel case

10

look like this

11

look like this

12

point out

1

unapologetically

2

consistency

3

importantly

4

inconsistent

5

judgment

6

one at a time

7

one at a time

8

one at a time

9

one at a time

10

one at a time

11

one at a time

12

one at a time

```

1 # This program says hello
  and asks for my name.
2 print('Hello, world!')
  print('What is your name?')
3 myName = input() # ask for their name
4 print('It is good to meet
  you, ' + myName)
5 print('The length of your
  name is:')
  print(len(myName))
6 print('What is your age?')
  # ask for their age.
  myAge = input()
  print('You will be ' + str(int(myAge)
  + 1)
  ' in a year.')

Mu
# hello.py

```

```

1 PasswordFile = open('Secret
  PasswordFile.txt')
2 secretPassword = PasswordFile.
  read()
3 print('Enter your Password')
  typedPassword = input()
4 if typedPassword == secretPass
  word:
5 print('Access Granted')
6 if typedPassword ==
  '12345':

```

unexpectedly من المفاجيء

8 plenty الكثير

9 المعادلات الرياضية بالرياضيات

10 $2 \times 3 \Rightarrow 6$ قوة $\rightarrow$

11 $22 \div 8 \Rightarrow 2.75$ الباقي $\rightarrow$

12 $22 \div 8 \Rightarrow 2$ القسمة $\rightarrow$

13 $3 \times 5 \Rightarrow 15$ ضرب $\rightarrow$

14 $5 - 2 \Rightarrow 3$ طرح $\rightarrow$

15 $2 + 2 \Rightarrow 4$ جمع $\rightarrow$

precedence الأسبقية

convention عرف

3 أسبقية أو ترتيب القاطير الرياضية

- 4 (من اليسار لليمين) $**$ ①
(من اليمين لليسار) $+$ ②
(من اليمين لليسار) $-$ ③

6 الأرقام تصير ترتيبا لعمليته
7 * تذكر ودرسه سابقا في كتاب الرياضيات
8 المتكلم محمد الفرسية

PEMDAS

8 طرح . جمع . قوة . قسمة

$2 + 3 \times 6 \Rightarrow 20$
 $(2 + 3) \times 6 \Rightarrow 30$

$4856878 \times 578453 \Rightarrow$

28093077826734

$2 \times 8 \Rightarrow 256$

$23 / 7 \Rightarrow 3.28571428571$

$23 // 7 \Rightarrow 3$

$23 \% 7 \Rightarrow 2$

$2 + 2 \Rightarrow 4$
 $(5 - 1) \times ((7 + 1) / (3 - 1)) \Rightarrow 16.0$

ترتيب عمل الرياضيات بأخرى
 $(5 - 1) \times ((7 + 1) / (3 - 1))$
 $4 \times ((7 + 1) / (3 - 1))$
 $4 \times (8 / 2)$
 4×4
 16.0

Fundamental الأساس
worst الأسوأ

integer عدد صحيح

42 عدد صحيح
42.0 Floats (Floating point)

$3.14 \rightarrow \pi$

- linux-ubuntu

8 sudo apt-get install python3

4 < < < idle3

9 < < < python3-pip

10

ننزل Mu

https://codewith.mu/

11

terminal في ubuntu

python3 -m mu

عندما تنزل mu فمعرض

البرامج Select Mode وهي

1 Adafruit CircuitPython

2 BBC, micro:bit

Pygame Zero

3 python 3

نحنا - python3

4 x تم اعلان تنزيل Mu اصلا

١٠٥٠٠ او ما بعدة زبائن من

5 تنصيب مودولات (طرق اخرى)

6 - Schoolroom الفصل الصف المدرسي

7 - Communities المجتمعات اشخاصك - شارك

8 folks الناس
indeed في الحقيقة
آلاف

IDLE =

integrated development
and learning Environmentننقله في ubuntu
terminal - نكتب في
idle3 >البرنامج الذي يتألف من نظام
الامر لتسهيل الامر.تفاعلي interactive
مترجم - فسر interpreter
لوحة - طرفيات Paneالمدارات - Prompts
الحالة (المضخة)

مخترع invented

عند كتابة shell في interactive
(نافذة التفاعل) مائلي>>> print('Hello, world!')
Hello, world!

تدري مودولات طرفيات

في ويندوز نستخدم PIP
في لينوكس نستخدم pipمعتاد accustomed
محاضرة - lectures
محاضر

Friday
Vendredi

14

January - Janvier

الجمعة ١٤

كانون الثاني - يناير

11 Jumada-2 6 /1443

Day: 14 / 351

2022

Week2, Month1

١١ جمادى الثاني ١٤٤٣/٦

| | |
|--------------------------------|---------------------------------|
| probably من المحتمل | wizard الساحر |
| 8 on purpose عن قصد | Arcane سري |
| implicit ضمنيًا، بغير صراحة | t tedious ملل |
| 9 traceback | wand عصا السحر |
| quotes الأقوال | incredible مذهبي |
| 10 tons أطنان | feats مفاخر - مناقب |
| forum منتدى | |
| 11 Stackoverflow.com | |
| reddit.com/r/learnprogramming | |
| 12 | interactive shell |
| نسخ الكود في الموقع التالي | REPL (Read-Evaluate-Print-Loop) |
| 1 pastebin.com | فوريًا - فوريًا |
| أف gist.github.com | مات |
| 2 * Came up | expression تعبير |
| etiquette السلوك | 777 2 + 2 4 |
| 3 unreasonable غير معقول | Value Operators |
| demands مطالب | 4 |
| 4 discover اكتشف | Evaluate تقييم |
| appliance جهاز - آلة | evaluate down تنقص |
| 5 along the way، | * تعبر القيمة لقيمة أخرى |
| amateurs الهواة | عبارة عن تغيير رقم |
| 6 beyond its scope. | 777 2 4 |
| invaluable ثمين - نفيس | 2 |
| 7 Journey رحلة | |
| Syntactical نحوي | |
| 8 constructions أنشآت - بنايات | |
| handy مفيد | |

Automate boring stuff مفردات مملوك

| | | |
|----|-----------------|---------------------|
| 8 | Figure out | experiment |
| | breaking down | intelligently |
| 9 | debugging | Functions (الوحدات) |
| | folks | instructs |
| 10 | gasp | manageable |
| | Capable | chunks |
| 11 | Contribute to | mysterious |
| | myth | efficiency |
| 12 | grade school | point out |
| | on top of that, | Syntax |
| 1 | dozen | Show up |
| | mindset | Comprehensions |
| 2 | knitting | misleading |
| | Canvas | stuff |
| 3 | yarn | invaluable |
| | bricks | Foundation |
| 4 | constraints | the gang |
| | endless | For not minding |
| 5 | raw | cheers |
| | infinite | roommate |
| 6 | knit | retail |
| | sweater | employees |
| 7 | automate | spreadsheet |
| | rundown | stack |
| 8 | expressions | competitors |
| | تعبير | |

Wednesday 12
Mercredi

الأربعاء ١٢

January - Janvier

كانون الثاني - يناير

9 Jumada-2 6/1443

Day: 12 / 353

2022

Week2, Month1

9 جمادى الثاني ١٤٤٣/٦

| | | | |
|----|--------------------|--|------------------|
| | Printouts | | involve |
| 8 | scattered | | interacting with |
| | affection | | skyrocketed |
| 9 | stacked | | affectionately. |
| | luggage | | pythonistas |
| 10 | idiot | | serpentine |
| | reasonable | | anxiety. |
| 11 | glowing | | notion |
| | furiously | | arithmetic. |
| 12 | cryptic | | boot camps |
| | streams | | ambitious |
| 1 | veteran | | rockstar |
| | sophisticated | | trivial |
| 2 | competitor | | Armed |
| | swiss Army knife. | | conventions |
| 3 | crunch | | trade off |
| | intricate | | throwaway. |
| 4 | surreal | | — |
| | monthly | | structuring |
| 5 | deductions | | — |
| | configure | | various |
| 6 | countless | | parse |
| | unaware | | web scraping |
| 7 | unaware | | programmatically |
| | whom | | spreadsheet |
| 8 | | | behalf |

**** Before the for loop ****

**** A new loop iteration ****

value of num: 0

value of tot: 0

**** A new loop iteration ****

value of num: 1

value of tot: 1

**** A new loop iteration ****

value of num: 2

value of tot: 3

(keeping Track of your iterator variable and iterable)

* what is the type of your iterable?

n = ["word", "phrase", 8, ("beam")]

For item in n:

print(item)

⇒ List

* what is the type of your iterable?

t = "Couch"

For z in t

print(z)

⇒ String

y = 18

For z in y:

print(z)

⇒ Error

Python is unable to iterate over integers and floats

Printing Intermediate Results

8

w = range(10)

9

tot = 0

10

for num in w:

tot += num

11

print(tot)

⇒ 45

12

w = range(10)

1

tot = 0

2

for num in w:

print(num)

3

tot += num

print(tot)

4

print(tot)

0
1
1
2
3
3
4
4
5
5
6
6
7
7
8
!8
AUG

5

w = range(10)

tot = 0

6

print("**** Before the for Loop ****")

for num in w:

7

print("**** A New Loop Iteration ****")

print("value of num: ", num)

tot += num

8

print("value of tot: ", tot)

print("**** End of for Loop ****")

print("Final total: ", tot)

Ewir ✓

```

8  nums = range(1, 11)
   accum = 0
9  For w in nums:
   accum = w accum + w
10 print(accum)

```

⇒ 55

```

11 nums = range(3, 11)
   accum = 0
12 For w in nums:
   accum = accum + w
1  print(accum)

```

⇒ 52

Naming variables in For Loops

X is a list defined elsewhere

```

4  For x in y:
   print(x)

```

genres is a list elsewhere

```

6  For genre in genres:
   print(genre)

```

```

8  For Fruit in Fruits:
   print(Fruit)

```

Accumulator
variable

The accumulator pattern

8 `nums = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]``accum = 0`9 iterator For `w` in `nums`:`accum = accum + w`10 `print(accum)`

11

| | |
|--------------------|------------------------|
| <code>accum</code> | ← accumulator variable |
| <code>w</code> | ← iterator variable |

12

1 `nums = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]`2 `print("range(5): ")`3 `for i in range(5):`
`print(i)`

⇒

| |
|-----------|
| range(5): |
| 0 |
| 1 |
| 2 |
| 3 |
| 4 |

4 `print("range(0, 5): ")`5 `for i in range(0, 5):`
`print(i)`

⇒

| |
|--------------|
| range(0, 5): |
| 0 |
| 1 |
| 2 |
| 3 |
| 4 |

Notice the casting of 'range' to the 'list'

6 `print(list(range(5)))`⇒ `[0, 1, 2, 3, 4]`7 `print(list(range(0, 5)))`⇒ `[0, 1, 2, 3, 4]`

Note: 'range' function is already as 'list' in the text that

8 `print(range(5))`⇒ `[0, 1, 2, 3, 4]`

تمرين ٣: أطول قائمة أرقام 67 ← 0

```
8 nums = []
   for x in range(68):
9       nums.append(x)
```

```
10 print(nums)
```

↓

```
11 [0, 1, 2, 3, 4, 5, 6, ..., 66, 67]
```

تمرين ٤: لدينا سلسلة - أنشئ قائمة تحتوي على عدد حروف كل كلمة في (2) لفظ

```
1 original_str = "The quick brown rhino jumped over the..."
   str_list = original_str.split()
2 num_words_list = []
   for x in str_list:
3       num_words_list.append(len(x))
```

```
4 print(num_words_list)
```

↓

```
5 [3, 5, 5, 5, 6, 4, 3, 9, 4, 3]
```

تمرين 5: أنشئ - لفظة فارغة لتتزايد بها حرف 'b' في 7 مرات

```
6 lett = ""
7 for x in range(7):
   lett = lett + "b"
```

```
8
```

How many times will loop iterate =

P = [3, 4, "Me", 3, [], "why", 0, "Tell", 9.3]

8 For ch in P:

print(ch)

9

9

10 9 = P[8] (9.3)

11

المطلوب جمع الأرقام الواردة في السلسلة

addition_str = "2+5+10+20"

12 addition_list = addition_str.split("+")

sum_val = 0

1 For x in addition_list:

sum_val += int(x)

2

print(sum_val)

3

المطلوب حساب متوسط درجات الطلاب في مادة الرياضيات

4

week_temps_f = "75.1, 77.7, 83.2, 82.5, 81.0, 79.5, 85.7"

5 week_temp_list = week_temps_f.split(",")

week_temp_sum = 0

6 For x in week_temp_list:

week_temp_sum += float(x)

7

avg_temp = week_temp_sum / (len(week_temp_list))

8

Friday
Vendredi

26

٢٦ الجمعة

August - Aout

آب - أغسطس

28 Muharram 1 / 1444

Day: 238 / 127

2022

Week34, Month8

٢٨ محرم ١٤٤٤/١

import turtle # set up alex

8

wn = turtle.Screen()

alex = turtle.Turtle()

9

for acolor in ["yellow", "red", "purple", "blue"]:

10

alex.forward(50)

alex.left(90)

repeat
four
times

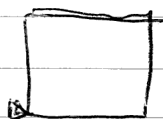
11

wn.exitonclick()

12



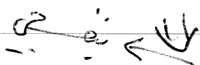
1



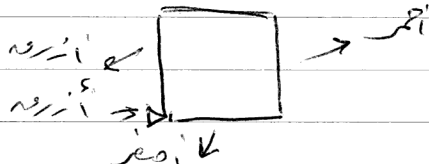
2

alex.color(acolor) *تغيير اللون*

3



4



5

6

* How does python know what statements are contained in the loop body?

7

They are ~~more~~ indented to the same degree from the loop header.

8

Thursday 25
Jeudi

August - Aout

الخميس ٢٥

آب - أغسطس

27 Muharram 1 / 1444

Day: 237 / 128

2022

Week34, Month8

٢٧ محرم ١٤٤٤/١

Lists and Loops

8 import turtle # Set up alex

wn = turtle.Screen()

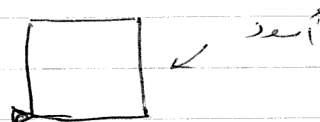
9 alex = turtle.Turtle()

10 For i in [0,1,2,3]: #repeat four times

alex.forward(50)

11 alex.left(90)

12 wn.exitonclick()



1 Fruits = ["apple", "orange", "banana", "cherry"]

2 For afruit in fruits: #by item

print(afruit)

3

4

5

6

7

8

apple
orange
banana
cherry

8
AUG

Wednesday 24
Mercredi

August - Aout

الأربعاء ٢٤

آب - أغسطس

26 Muharram 1 / 1444

Day: 236 / 129

2022

Week 34, Month 8

٢٦ محرم ١٤٤٤

For achar in "Go spot Go":

8 print(achar)

↓

9 G

O

10 فرغ →

S

11 P

O

12 T

فرغ →

1 G

O

2

3 * How many times is the word HELLO printed by the following statements?

S = "python rocks"

4 For ch in S[3:8]:

print("HELLO")

5

↓

5

لن يـد صـرـوـف (non r) = 5 صـرـوـف

7

8

Tuesday
Mardi

23

الثلاثاء ٢٣

August - Aout

آب - أغسطس

25 Muharram 1 / 1444

Day: 235 / 130

2022

Week 34, Month 8

٢٥ محرم ١٤٤٤/١

```
original_str = "The quick brown rhino jumped over --"  
8 num_chars = 0  
for i in original_str:  
9     num_chars += 1
```

10 The Loop

```
For name in ["Joe", "Amy", "Brad", "Angelina", "Zuki",  
11 "Thandi", "Paris"]
```

```
Print("Hi", name, "please come to my party on Saturday")  
12
```

Hi Joe please come to my party on Saturday!

1 Hi Amy < < < < < < <

< Brad < < < < < < <

2 < Angelina < < < < < < <

< Zuki < < < < < < <

3 < Thandi < < < < < < <

< Paris < < < < < < <

4 Strings and Loops

* How many times is the word HELLO printed by
5 the following statements?

S = "python rocks"

6 For ch in S:

Print(HELLO)

7

١٢ = python rocks
٨

8
AUG

Monday
Lundi

22

الاثنين ٢٢

August - Aout

آب - أغسطس

24 Muharram 1 / 1444

Day: 234 / 131

2022

Week 34, Month 8

٢٤ محرم ١٤٤٤/١

Several-things = ["Hello", 2, 4, 6.0, 7.5, 234352354, "the end",
"u", 99]

For i in several-things:

print(i)

For x in several-things:

print(type(x))



Hello

2

4

6.0

7.5

234352354

the end

99



<class 'str'>

<class 'int'>

<class 'int'>

<class 'float'>

<class 'float'>

<class 'int'>

<class 'str'>

<class 'str'>

<class 'int'>

Sunday
Dimanche

21

الإحد ٢١

August - Aout

آب - أغسطس

23 Muharram 1/1444

Day: 233 / 132

2022

Week34, Month8

محرم ٢٣ ١٤٤٤/١

8 l = ['w', '7', '0', '9']
m = l[1] ⇒ String النوع

9
b = "My, what a lovely day"
10 x = b.split(',') ⇒ list النوع
z = " ".join(x)

11 y = z.split()
a = " ".join(y) ⇒ String النوع

12
my_str = "MICHIGAN"
1 for i in my_str:
print(i)

2 ⇒ M
I
c
H
I
G
A
N

5 str_list = ["Hello", "", "goodbye", ...]
For x in str_list:
6 print(len(x))

7 ⇒ 5
0
7
9
13

8
AUG


```

8 wds = ["red", "blue", "green"]
  glue = ";"
  s = glue.join(wds)
9  print(s)           => red;blue;green
  print(wds)          => ['red', 'blue', 'green']
10
  print('***'.join(wds)) => red***blue***green
11  print("").join(wds)  => redbluegreen
  Create a variable (output) and assign it to a list
12  whose elements are the words in the string (str1).

1  str1 = "OH THE PLACES YOU'LL GO"
2
  output = str1.split()

3  create a variable called (wrds) and assign to it a list
  whose elements are the words in the string (sent). Don't worry
4  about punctuation.

5  sent = "The bicentennial for our university is in 2017!"
  wrds = sent.split()
6
  let = "z"
7  let_two = "p"
  c = let_two + let
8  m = c * 5
  print(m) => pzpzpzpzpz

```

```

8 seasons = ["winter", "spring", "summer", "fall"]
9 print(seasons.index("autum")) =>
10                                     (ValueError:
                                     list.index(x):
                                     x not in list on
                                     line 3)

```

```

11 qu = "wow, welcome week!"
12 ty = qu.index("we") => 6
    ty = qu.index("wer") => 5 (There is an error)

```

Splitting and Joining Strings

```

1 song = "The rain in Spain..."
2 wds = song.split()
3 print(wds) => ['The', 'rain', 'in', 'Spain...']
4 wds = song.split('ai')
5 print(wds) => ['The', 'n in sp', 'n...']

```

```

6 'leaders and best'.split("e")

```

```

7 => ['l', 'ad', 'rs and b', 'st']
8

```

Count and index

8

a = "I have had an apple on my desk before!"

9

print(a.count("e")) $\Rightarrow$ 5

10

print(a.count("ha")) $\Rightarrow$ 2

11

z = ['atoms', 4, 'neutron', 6, 'proton', 4, 'electron', 4, 'atoms']

12

print(z.count("4")) $\Rightarrow$ 4

print(z.count(4)) $\Rightarrow$ 3

1

print(z.count("a")) $\Rightarrow$ (2) ? سؤال

print(z.count("electron")) $\Rightarrow$ 1

2

music = "Pull out your music and dancing can begin"

3

bio = ["Metatarsal", "Metatarsal", "Fibula", [], "Tibia", "Tibia", 43, "Femur", "occipital", "Metatarsal"]

4

print(music.index("a")) $\Rightarrow$ 14

5

print(music.index("Your")) $\Rightarrow$ 9

6

print(bio.index("Metatarsal")) $\Rightarrow$ 0

print(bio.index([])) $\Rightarrow$ 3

7

print(bio.index(43)) $\Rightarrow$ 6

8

Wednesday 17
Mercredi

August - Aout

الأربعاء ١٧

آب - أغسطس

19 Muharram 1/1444

Day: 229 / 136

2022

Week33, Month8

19 محرم ١٤٤٤/١

Print((Fruit + [1]) * 4) =>

8

9
10
['apple', 'orange', 'banana', 'cherry', 1,
'apple', 'orange', 'banana', 'cherry', 1,
'apple', 'orange', 'banana', 'cherry', 1,
'apple', 'orange', 'banana', 'cherry', 1]

11
12
print([0,1]*4) => [0,1,0,1,0,1,0,1]
print([0]*4) => [0,0,0,0]

fruit = ['apple', 'orange', 'banana', 'cherry']

1

numlist = [6,7]

2

newlist = fruit + numlist => ['apple', 'orange',
'banana', 'cherry',
6,7]

3

a list = [1,3,5]

4

b list = [2,4,6]

Print(a list + b list) => [1,3,5,2,4,6]

5

Print(a list * 3) => [1,3,5,1,3,5,1,3,5]

6

7

8

8
AUG

The Slice operator

```

8      ⑧      ①      ②      ③      ④
julia = ("Julia", "Roberts", 1967, "Duplicity", 2009,
9      ⑤      ⑥
        "Actress", "Atlanta, Georgia")

10  print(julia[2]) ⇒ 1967
    print(julia[2:6]) ⇒ (1967, 'Duplicity', 2009, 'Actress')
    print(len(julia)) ⇒ 7
11  julia = julia[:3] + ("Eat Pray Love", 2010) + julia[5:]
    print(julia) ⇒ ('Julia', 'Roberts', 1967, 'Eat Pray Love',
12                2010, 'Actress', 'Atlanta, Georgia')

```

```

1  s = "python rocks"
    print(s[3:8]) ⇒ hon r

```

```

2      ⑥      ①      ②      ③      ④      ⑤      ⑥
alist = [3, 67, "cat", [56, 57, "dog"], [3.14, False]]
3  print(alist[4:]) ⇒ [[3.14, False]]

```

Concatination and Repetition

```

5  Fruit = ["apple", "orange", "banana", "cherry"]
    print([1, 2] + [3, 4]) ⇒ [1, 2, 3, 4]
6  print(Fruit + [6, 7, 8, 9]) ⇒ ['apple', 'orange',
    'banana', 'cherry', 6, 7, 8, 9]
7  print(Fruit * 4) ⇒ ['apple', 'orange', 'banana',
    'cherry', 'apple', 'orange', 'banana',
    'cherry', 'apple', 'orange', 'banana',
    'cherry']
8

```

```
mylist = []
```

```
mytuple = ()
```

```
print(type(mylist)) ⇒ <class 'list'>
```

```
print(type(mytuple)) ⇒ <class 'tuple'>
```

index operator : working with the characters of a string

```
s = "python"
```

```
print(s[0]) → p
```

```
print(s) → python
```

| 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| p | y | t | h | o | n |
| 6 | 5 | 4 | 3 | 2 | 1 |

```
s = "python"
```

```
mylist = ["one", 2, "three"]
```

```
print(mylist) → ['one', 2, 'three']
```

```
print(mylist[1]) → 2
```

```
print(mylist[2]) → 'three'
```

```
print(s[len(s) - 1]) ⇒ n
```

```
print(len(mylist)) → 3
```

```
print(len(s)) → 6
```

```
print(mylist[0]) → 'one'
```

```
print(s[0]) → p
```

```
print(mylist[-1]) → 'three'
```

```
print(s[-3]) ⇒ h
```

```
print(s[-1]) ⇒ n
```

Sunday 14
Dimanche

August - Aout

الأحد ١٤

آب - أغسطس

16 Muharram 1 / 1444

Day: 226 / 139

2022

Week 33, Month 8

١٦ محرم ١٤٤٤

s = '100'

8

mylist = [100]

i = 100

9

Print(mylist + 5) ⇒ TypeError: can only concatenate
list to list on line 5

10

Tuples

11

mylist = [100]

mytuple = (100)

12

Print(type(mylist)) → <class 'list'>

Print(type(mytuple)) → <class 'int'>

1

mylist = ["one", 2, "three"]

2

mytuple = ("one", 2, "three")

3

Print(type(mylist)) ⇒ <class 'list'>

Print(type(mytuple)) ⇒ <class 'tuple'>

4

mylist = [100]

5

mytuple = (100)

mytuple2 = (100,)

6

Print(type(mylist)) → <class 'list'>

7

Print(type(mytuple)) → <class 'int'>

Print(type(mytuple2)) → <class 'tuple'>

8

8 `print(s) ⇒ 5`
`print(i+10) ⇒ 15`
`print(s+10) ⇒ TypeError: Cannot concatenate 'str'`
`and 'int' object on line 4`
`print(int(s)+10) ⇒ 15`
`print(i+10) ⇒ 15`

11 `s = '5 hundred'`
`i = 5`
`print(float(s)+10) ⇒ ValueError: float: argument`
`5 hundred is not number on`
`line 4`

1 `print(i+10)`

2

8
AUG

3 `s = '5'`
`i = 5` 5.0
`print(float(s)+10) → 15.0`
`print(i+10) → 15`

5 `s = '5 hundred'`
`myList = ["one", 2, "three"]`
`print(myList) → ['one', 2, 'three']`

7

8

Strings and Lists

s = 'Hello world'

m = ('('))

this is a

multi-line

string!

'''

print(m)

⇒

This is a
multi-line
string!

print(s) ⇒ Hello world

وعلنا الوضع في السطر عينة آفيس من زودعة وبع لنهاية عينة
آفيس نكرة ← خطأ من

s = ('Hello world')

print(s)

⇒ SyntaxError: Bad token on line 1

s = '5'

i = 5

print(type(s)) ⇒ <class 'str'>

print(type(i)) ⇒ <class 'int'>

Common Errors

8

Syntax Error

alex.forward(50) ⇒ SyntaxError: bad input
on line 6

9

alex = Turtle.turtle()

10

alex = Turtle.turtle() ⇒ NameError: name
'Turtle' is not defined
on line 3

11

* Jane = turtle.Turtle()

12

right.jane(90) ⇒ NameError: name 'right'
is not defined on line 6

1

jane

2

jane.right(90)

8
AUG

3

import turtle

wn = turtle.Screen()

4

jane = turtle.Turtle()

5

for _ in range():

jane.color("green", "yellow")

6

jane.forward("50")

jane.right(90)

7

TypeError: range() takes
at least 1 arguments
(0 given) on line 5

For - in range(3):

8

أولاً

```

{ bob.right(135)
  bob.forward(50)
  bob.right(90)
  bob.forward(50)
    
```



9

10

11

ثانياً

```

bob.right(45)
bob.forward(50)
    
```

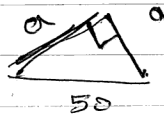


12

1

١٣٥ ← ٤٥ → ٩٠

2



نريد كل الزوايا لعديت

3

$$a^2 + a^2 = 50^2$$

4

```

import turtle
import math
wn = turtle.Screen()
bob = turtle.Turtle()
bob.right(90)
bob.forward(50)
bob.left(90)
bob.forward(50)
bob.left(90)
bob.forward(50)
bob.left(90)
    
```

5

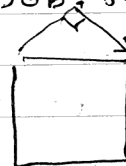
6

7

8

```

bob.forward(50)
bob.right(135)
dist = math.sqrt(50*50/2)
bob.forward(dist)
bob.right(90)
bob.forward(dist)
    
```



(Know your Error Messages) Errors

8 syntax errors → Print ("Hello world!")
نقص أو زيادة في علامات الاقتباس

9 syntaxError: bad input on Line 3
syntaxError: EOF in multi-line statement
on line 3

10 Print(3/0) → (Runtime Error)
11 ZeroDivisionError: integer division
or module by zero on line 1

12 runtime error → Division by zero

1 syntax error → Forgetting a semi-colon at
the end of a statement
2 where one is required.

3 semantic error → Forgetting to divide by 100
when printing a percentage amount

incremented programming

5 import turtle
wn = turtle.Screen()
6 bob = turtle.Turtle()
bob.right(90)
7 bob.forward(50)
bob.left(90)
8 bob.forward(50)
bob.left(90)
bob.forward(50)

عبر
bob.left(90)
bob.forward(50)



Swi-Cute!

8

9

10

11

12

1

2

3

4

5

6

7

8

variables

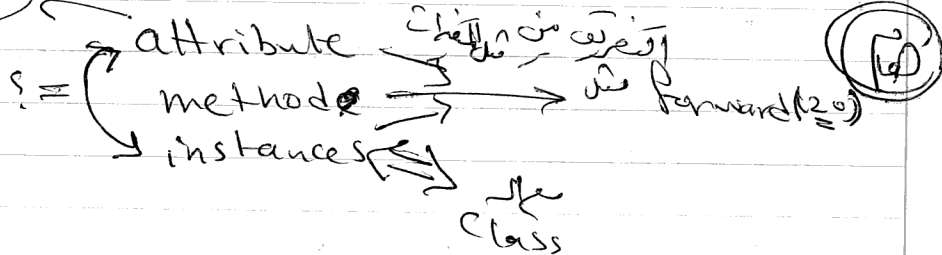
statements

expression

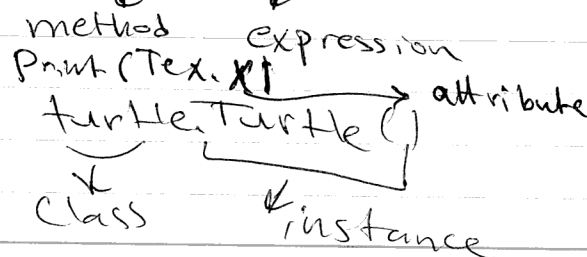
500
3.14
200 + 300
10.0 + 5.0

len → Function
Square
Sub

متغير
Variable



Tex.forward(10 + 10)



Importing modules

① the random module

import random

معي

prob = random.random()
print(prob)

⇒ 0.0392524 -

diceThrow = random.randrange(1, 7)
print(diceThrow)

returning
int
one of
1, 2, 3, 4, 5, 6

⇒ 3

From random import randrange, random

prob = random()
print(prob)

diceThrow = randrange(1, 7)
print(diceThrow)

Saturday
Samedi

6

August - Aout

6 السبت

آب - أغسطس

8 Muharram 1 / 1444

Day: 218 / 147

2022

Week32, Month8

محرم ٨ / ١٤٤٤

import turtle

8

wn = turtle.Screen()

wn.bgcolor("lightgreen")

9

tess = turtle.Turtle()

tess.color("blue")

10

tess.shape("turtle")

11

dist = 5

tess.up()

12

for i in range(30):

tess.stamp()

1

tess.forward(dist)

tess.right(24)

2

dist = dist + 2

wn.exitonclick()

3

4

5

6

7

8

* Start with size 5 and
grow by 2
* leave an impression on the
Canvas
* move tess along
* and turn her.



Friday
Vendredi

5

٥ الجمعة

August - Aout

آب - أغسطس

7 Muharram 1 / 1444

Day: 217 / 148

2022

Week 31, Month 8

٧ محرم ١٤٤٤

import turtle

8

wn = turtle.Screen()

9

elan = turtle.Turtle()

10

distance = 50

angle = 90

11

for _ in range(10):

elan.forward(distance)

12

elan.right(angle)

distance = distance + 10

1

angle = angle - 3

2

3

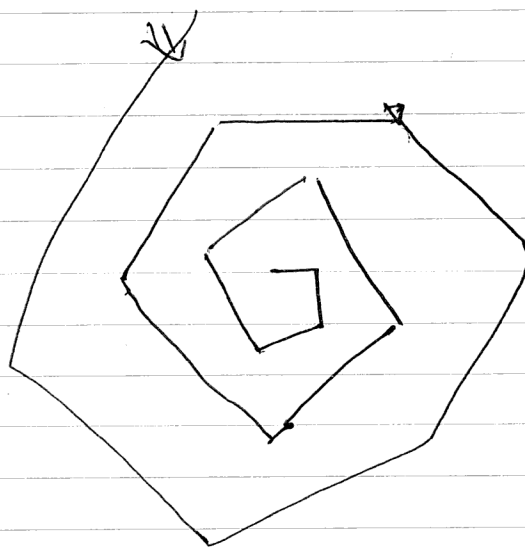
4

5

6

7

8



8
AUG

Thursday
Jeudi

4

August - Aout

الخميس ٤

آب - أغسطس

6 Muharram 1 / 1444

Day: 216 / 149

2022

Week 31, Month 8

٦ محرم ١٤٤٤

import turtle

8

wn = turtle.Screen()

9

elan = turtle.Turtle()

10

distance = 50

for _ in range(6):

11

elan.forward(distance)

elan.right(90)

12

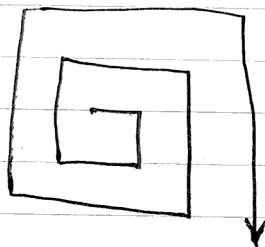
distance = distance + 10

1



2

3



4

5

6

7

8

Wednesday
Mercredi

3

الأربعاء ٣

August - Aout

آب - أغسطس

5 Muharram 1 / 1444

Day: 215 / 150

2022

Week 31, Month 8

محرم ١ / ١٤٤٤ هـ

Repetition with a for loop

print("This will execute first")

for - in range(3):

print("This ^{line} will execute three times")

print("This line will also execute three times")

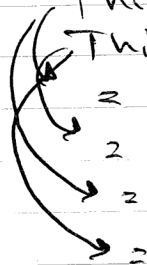
print("Now we are outside of the for loop!")



This will execute first

This ^{line} will execute three times

This line will also execute three times



Now we are outside of the for loop!

8
AUG

The Accumulator pattern with strings

8

`s = input("Enter some text")`

9

`ac = ""``for c in s:`

10

`ac = ac + c + "-" + c + "-"`

11

`print(ac)`

12

`s → "cat" → c-c-a-a-t-t-`

1

`byzo = 'Hello world!'`

2

`c = 0``for (x) in byzo:`

3

`z = x + "!"``print(z)``c = c + 1`

4

byzo explains is to C ← accumulator var.
X ← iterator(loop) var.

5

`cawdra = ['candy', 'daisy', 'pear', '']`

6

`t = 0``for item in cawdra:`

7

`t = t + len(item)`

8

`cawdra` ← sequence

2022

8 [9] أكتب كود ديف 'horseback riding' في الموقع 2

9 sports = ['cricket', 'Football', ...]
print(sport.insert(2, 'horseback riding'))

10 London في قائمة السفر في القائمة [5]

11 trav_dest = ['Beirut', 'Milan', 'pittsburgh', 'Buenos Aires',
'London', '...', '...']

12 trav_dest.remove('London')

1 The Accumulator pattern with Lists

2 nums = [3, 5, 8]

accum = []

3 for w in nums:

x = w * 2

4 accum.append(x)

print(accum) ⇒ [9, 25, 64]

5

nums → [3 | 5 | 8]

6

accum → [9 | 25 | 64]

7

8

Friday
Vendredi

30

الجمعة ٣٠

September - Septembre

أيلول - سبتمبر

4 Rabie-1 3 / 1444

Day: 273 / 92

2022

Week39, Month9

٤ ربيع الأول ١٤٤٤/٣

٢٣. لنناقش كل عنصر من عناصر هذه السلسلة بحري اسم العنصر

8 (المبيع) - مكتبة دروس كود

أطوب كتاب كود يوتي في الزفير

9 The store has 12 shoes, each for 29.99 USD

10 (Format method) . Format string

11 inventory = ["shoes, 12, 29.99", "Shirts, 20, 9.99",
"sweatpants, 25, 15.00", "scarves, 13, 7.75"]

12 for temp in inventory:

temp = temp.split(",")

1 item = temp[0]

quantity = temp[1]

2 amount = temp[2]

str1 = "The store has {} {}, each for {} USD."

3 str1 = str1.format(quantity, item, amount)

4 print(str1)

⇒ The store has 12 shoes, each for 29.99 USD.

5 < < < 20 shirts, < < 9.99 < .

< < < 25 sweatpants < , 15.00 < .

6 < < < 13 scarves < , 7.75 < .

7

8

Thursday 29
Jeudi

September - Septembre

الخميس ٢٩

أيلول - سبتمبر

3 Rabie-1 3 / 1444

Day: 272 / 93

Week 39, Month 9

3 ربيع الأول ١٤٤٤ / ٣

۱۳] فیما یلی هو مجموعہ عدداتی لطافہ الطیر کے (مضی) سامع . آئینہ کو
میرے عدد عدداتی آئینہ کے . ا. او آکر اوضع انسیت ہے (تجربہ
a-score.

9 scores = "6780907893207989969792887968589098"
scores_list = scores.split()

10 a_score = 0

For score in scores_list:

11 if int(score) >= 90:

a_score += 1

12 print(a_score)

⇒ 10

۱-phrase قلبیہ ہے p-phrase و غرض ہے

2 p-phrase = "was it a car or a cat I saw"

r-phrase = ""

3 for i in range(len(p-phrase)):

r-phrase = r-phrase + p-phrase[(len(p-phrase)-1)-i]

4

print(r-phrase) ⇒ w I tac a ro rac a ti Saw

5

6

7

8

9
SEP

acronym

الكلمة المكونة من الحروف الأولى لكلمات في جملة

مثال: water, earth, air, fire → WEAIR

acronym

الكلمة المكونة من الحروف الأولى لكلمات في جملة

مثال: water, earth, air, fire → WEAIR

مثال: water, earth, air, fire → WEAIR

مثال: water, earth, air, fire → WEAIR

مثال: water, earth, air, fire → WEAIR

مثال: water, earth, air, fire → WEAIR

12 "HE. EW. WO"

1 stop words = ['to', 'a', 'for', 'by', 'an', 'am', 'the', 'so']

2 sent = "The water earth and air are vital"

3 sent_lower = sent.lower()

sent_lst = sent_lower.split()

4 acro = ""

For stopword in stop words:

5 if stopword in sent_lst:

sent_lst.remove(stopword)

6 For wrd in sent_lst:

if sent_lst.index(wrd) > (len(sent_lst) - 1):

7 acro = acro + wrd[0].upper() + wrd[1].upper()

else:

8 acro = acro + wrd[0].upper() + wrd[1].upper() + " "

print(acro) → [WA. EA. AI. AR. VI]

وهو المطلوب

Tuesday 27
Mardi

September - Septembre

الثلاثاء ٢٧

أيلول - سبتمبر

1 Rabie-1 3/1444

Day: 270 / 95

2022

Week39, Month9

1 ربيع الأول ١٤٤٤/٣

String format Method.

8

```
scores = [("Rodney Dangerfield", -1), ("Marlon Brando", 1),  
          ("you", 100)]
```

9

For person in scores:

10

```
    name = person[0]
```

```
    score = person[1]
```

11

```
    print("Hello " + name + " your score is " + str(score))
```

12

⇒ Hello Rodney Dangerfield. your score is -1

Hello Marlon Brando. your score is 1

1

Hello you. your score is 100

2

```
print("Hello {}. your score is {}".format(name,  
                                           score))
```

3

→ عايناه

4

```
origPrice = float(input('Enter the original price: $'))
```

5

```
discount = float(input('Enter discount percentage: %'))
```

```
newPrice = (1 - discount/100) * origPrice
```

6

```
calculation = '{}: {:.2f} discount by {}% is {:.2f}'.format(  
    origPrice, discount, newPrice)
```

7

```
print(calculation)
```

⇒

\$89.99 discounted by 20% is 71.99

71.9920

تعليق
على
Format
method

9
SEP


```
ss = " Hello, World "
```

```
8 els = ss.count("l")
```

```
print(els) → 3
```

٣ مرّة ل في ال string

```
print("###" + ss.strip() + "###") → ### Hello, world ###
```

10

```
news = ss.replace("o", "###")
```

```
11 print(news) → H##ll###, W##rrld
```

↑
الفراغات

↑
الفراغات

12

```
s = "python rocks"
```

```
1 print(s[1]*s.index("n")) → YYYXX
```

٥ * ٨

2

٨ هو ال index ل n على طول ال string (٨ هو ال index ل n)

٥ هو ال index ل * على طول ال string

3

```
4 name = "Rodney Dangerfield"
```

```
score = -1
```

```
5 print("Hello" + name + ". Your score is " + str(score))
```

6

```
⇒ Hello Rodney Dangerfield. Your score is -1
```

7

8

the identifier: 1406684 --- 968

8 origlist: [45, 32, 88, 'cat']

the identifier: 1406684 --- 376

9

origlist [45 | 32 | 88 | 'cat']
newlist [45 | 32 | 88 | 'cat']

10

11 origlist = [45, 32, 88]

aliaslist = origlist

12 origlist += ['cat']

origlist = origlist + ['cow']

1

origlist → [45 | 32 | 88 | 'cat']
aliaslist → [45 | 32 | 88 | 'cat']

2

3

non-mutating Methods on Strings

4

ss = "Hello, world"

5 print(ss.upper()) ⇒ "HELLO WORLD"

6 tt = ss.lower()

print(tt) ⇒ hello, world

7 print(ss) ⇒ Hello, world

8

Append vs. Concatenate

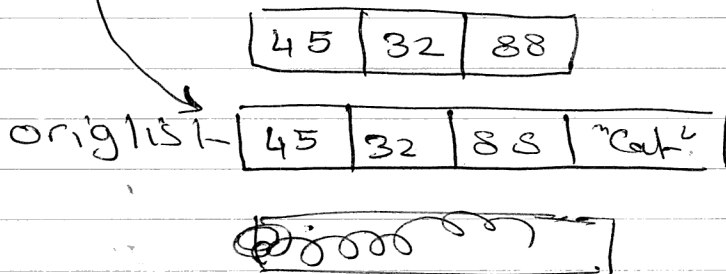
~~original~~

origlist = [45, 32, 88]

origlist.append("cat")

origlist = [45, 32, 88]

origlist = origlist + ["cat"]



origlist = [45, 32, 88]

print("origlist:", origlist)

print("the identifier:", id(origlist))

newlist = origlist + ["cat"]

print("newlist:", newlist)

print("The identifier:", id(newlist))

origlist.append("cat")

print("origlist:", origlist)

print("The identifier:", id(origlist))

id of list
before change

id of list
after change

id of list
after append

methods on Lists

8 1/ Mutating methods

9 ~~mylist.append(27)~~ mylist.append(5)
mylist.append(3)

mylist.append(12)

10 print(mylist) ⇒ [5, ~~27~~, 3, 12]
 و الله اعلم بالصواب

11 mylist.insert(1, 12) ⇒ mylist[1] = 12

print(mylist) ⇒ [5, 12, 27, 3, 12]

12 print(mylist.count(12)) ⇒ 2

1 print(mylist.index(3)) ⇒ 3

2 print(mylist.count(5)) ⇒ 1

mylist.reverse() ⇒ [12, 3, 27, 12, 5]

3 print(mylist) ⇒ [12, 3, 27, 12, 5]

4 mylist.sort() (3 5 12 12 27

print(mylist) ⇒ [3, ~~13~~, 12, 27]

5 del mylist[1]

mylist.remove(5)

6 print(mylist) ⇒ [3, 12, 12, ~~27~~]

7 lastitem = mylist.pop()

print(lastitem) ⇒ 27

8 print(mylist) ⇒ [3, 12, 12]

del mylist[1]
remove

del mylist[1]
mylist.remove(5)
result

Thursday 22
Jeudi

September - Septembre

الخميس ٢٢

أيلول - أيلول

26 Safar 2/1444

Day: 265 / 100

2022

Week38, Month9

٢٦ صفر ١٤٤٤/٢

sent = "The mall has excellent sales right now."

8 wrds = sent.split()

wrds[1] = "store"

9 new_sent = " ".join(wrds)

10

11 sent → "The mall has excellent sales right now."
"store"

12 wrds →

| | | | | | | |
|-----|------|-----|-----------|-------|-------|-----|
| The | mall | has | excellent | sales | right | now |
|-----|------|-----|-----------|-------|-------|-----|

1 new_sent → "The store has excellent sales right now."

2

عند كتابة آخر السطر = فائدة مع الكلمة زحل
أما إذا استمر السطر زحل (المفرد المفرد + كلمة) فائدة
لنقوم بعمل فحة

4

5

6

7

8

8 `lst = ['mercury', 'venus', 'earth', 'mars', 'jupiter', 'saturn', 'uranus', 'neptune', 'pluto']`

9 `lst.remove('pluto')`

10 `first_three = lst[:3]`

11

`lst` →

| | | | | | | | | |
|---------|-------|-------|------|---------|--------|--------|---------|-------|
| mercury | venus | earth | mars | jupiter | saturn | uranus | neptune | pluto |
|---------|-------|-------|------|---------|--------|--------|---------|-------|

12

`first_three` →

| | | |
|---------|-------|-------|
| mercury | venus | earth |
|---------|-------|-------|

13

ثم عند استخدام `remove` فإنها تفتقد عنصر القائمة الموجودة، ولا تقوم بحذفه.

2 `x = ["dogs", "cats", "birds", "reptiles"]`

3 `y = x`

4 `x += ['fish', 'horses']`

5 `y = y + ['sheep']`

6

`x` →

| | | | | | |
|------|------|-------|----------|------|--------|
| dogs | cats | birds | reptiles | fish | horses |
|------|------|-------|----------|------|--------|

7

`y` →

| | | | | | | |
|------|------|-------|----------|------|--------|-------|
| dogs | cats | birds | reptiles | fish | horses | sheep |
|------|------|-------|----------|------|--------|-------|

نفس الشيء - `obj + obj` - `obj + obj` يختلف عن `obj += obj`
 حيث أن `obj + obj` عبارة عن قائمة جديدة، فإننا نضعها في مكان جديد
 ونفس الشيء - `obj += obj` - `obj += obj`
 إضافة عنصرين `obj + obj` إلى `obj`.

هل هذه تسمى بـ alias confusion ؟

8

$b = [1, 9, 1, 1, 1]$

9

$z = b$

$b[1] = 1$

10

$z.remove(1)$

$print(z)$

11

الجواب نعم نعم الـ alias لأن مرجعية z هي نفس الـ b و تم التغير باستخدام كلا المتغيرات.

12

Yes, b and z reference the same list and changes are made using both aliases.

1

هل التسمية alias تسمى بـ confusion ؟

2

$sent = "Holidays can be a fun time when you have good company."$
 $phrase = sent$

3

$phrase = phrase + "Holidays can also be fun on your own!"$
(تغير)

4

الجواب No لأن الـ string غير قابلة للتغيير (immutable) فإن التسمية لن تكون متغيرة لأن الـ string غير قابل للتغيير.

5

$item = item + new_item$ مع المتغير $item$ فإنه يغير القيمة لأن الـ string غير قابل للتغيير. ولكن عند استخدام $+$ فإن ذلك لن يحدث.

No, Since a string is immutable, aliasing won't be as confusing. Beware of using something like $item = item + new_item$ with mutable objects though because it creates a new object. However, when we use $+=$ then that doesn't happen.

8

Aliasing

8 a = [81, 82, 83]

b = [81, 82, 83]

9 print(a is b) ⇒ False

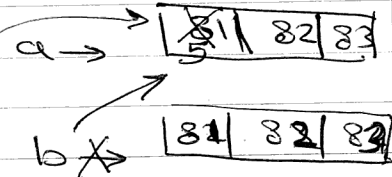
10 b = a

print(a == b) ⇒ True

11 print(a is b) ⇒ True

12 b[0] = 5

print(a) ⇒ [5, 82, 83]



Cloning Lists

2

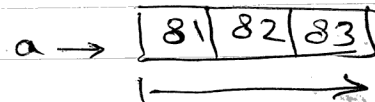
a = [81, 82, 83]

3

b = a[:] # make a clone using slice

4 print(a == b) ⇒ True

print(a is b) ⇒ False



5

b[0] = 5

6 print(a) ⇒ [81, 82, 83]

print(b) ⇒ [5, 82, 83]



7

8

List Element Deletion

8

`a = ['one', 'two', 'three']`

9

`del a[1]`

10

`print(a)` $\Rightarrow$ `['one', 'three']`

11

`alist = ['a', 'b', 'c', 'd', 'e', 'f']`

`del alist[1:5]`

`print(alist)`

$\Rightarrow$ `['a', 'f']`

12

objects and References

1

`a = "banana"`

2

`b = "banana"`

3

`print(a is b)` $\Rightarrow$ True

$a \rightarrow$ "banana"
 $b \rightarrow$

4

`a = [81, 82, 83]`

`b = [81, 82, 83]`

$a \rightarrow$ [81 | 82 | 83]

5

`print(a is b)` $\Rightarrow$ False

$b \rightarrow$ [81 | 82 | 83]

6

`print(a == b)` $\Rightarrow$ True

7

`print(id(a))` $\Rightarrow$ 3

`print(id(b))` $\Rightarrow$ 4

a و b متغيران
لهما نفس القيمة
ولكنهما متغيران مختلفان

8

Saturday 17
Samedi

September - Septembre

السبت ١٧

أيلول - سبتمبر

21 Safar

2/1444

260/105

2022

Week38, Month9

٢١ صفر ١٤٤٤/٢

المراد تفضل فاضلاً عن
(string) مراد تفضل فاضلاً عن

دُخِلَ دُونَ فُحْرًا سَتَاغِ الْفَيْضَ لِحْدِ صَبْرٍ

8

greeting = "Hello, world!"

9

newGreeting = 'J' + greeting[1:]

print(newGreeting)

⇒ Hello world!

10

print(greeting)

⇒ Hello world!

المراد

11

phrase = "Many moons"

phrase_expanded = phrase + "and many stars"

12

phrase_larger = phrase_expanded + "litter"

phrase_complete = "M" + phrase_larger[1:] + "the night sky"

1

excited_phrase_complete = phrase_complete[:-1] + "!"

2

⇓

3

phrase = "many moons"

phrase_expanded = "many moons and many stars"

4

phrase_larger = "many moons and many stars litter"

5

phrase_complete = "Many moons and many stars litter the night sky"

6

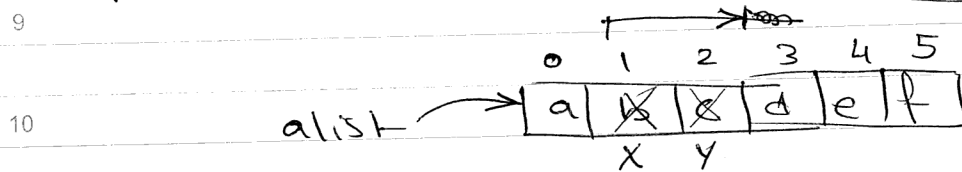
7

8

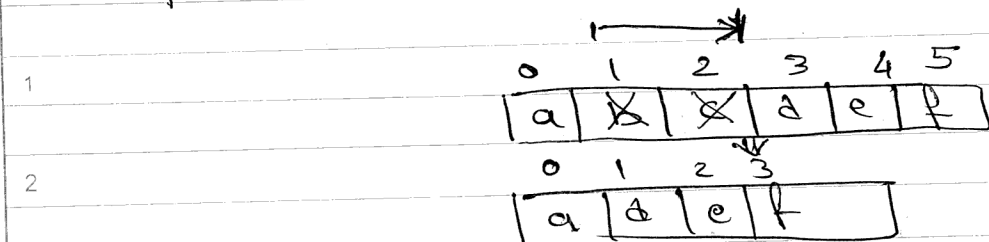
9

9 SEP

8 `alist = ['a', 'b', 'c', 'd', 'e', 'f']`
`alist[1:3] = ['x', 'y']`
`print(alist)` $\Rightarrow$ `['a', 'x', 'y', 'd', 'e', 'f']`



11 `alist = ['a', 'b', 'c', 'd', 'e', 'f']`
`alist[1:3] = []`
`print(alist)` $\Rightarrow$ `['a', 'd', 'e', 'f']`



3 `greeting = "Hello, world"`
`greeting[0] = 'J'` $\Rightarrow$ `# Error`
`print(greeting)`

نظراً
 TypeError: 'str' does not support item assignment

لا يمكن تغيير محتوى string
 item assignment

Thursday 15
Jeudi

September - Septembre

الخميس ١٥

أيلول - سبتمبر

19 Safar 2/1444

Day: 258 / 107

2022

Week37, Month9

19 صفر ١٤٤٤/٢

* اكتب كوداً يعد عدد الحروف المتحركة في كلمة s وقرن بينها وبين الحروف الساكنة
num-vowels: ذلك يعني ان الحروف المتحركة هي (a, o, i, e, u)
الحروف الساكنة: الحروف المتحركة في s

9

s = "singing in the rain and playing in the rain
are two - - -"

10

vowels = ['a', 'e', 'i', 'o', 'u']

11

num_vowels = 0

12

for char in s:

if char in vowels:

1

num_vowels += 1

print(num_vowels)

2

Sequence Mutation

3

* Mutability

4

Fruit = ["banana", "apple", "cherry"]

print(Fruit) ⇒ ['banana', 'apple', 'cherry']

5

Fruit[0] = "pear"

Fruit[-1] = "orange"

6

print(Fruit) ⇒ ['pear', 'apple', 'orange']

7

| | | |
|---------------------|---------|---------------------|
| "banana" | "apple" | "cherry" |
| "pear" | | "orange" |

8

0 1 2
3- 2- 1-

9
SEP

Wednesday 14
Mercredi

الأربعاء ١٤

September - Septembre

أيلول - سبتمبر

18 Safar 2/1444

Day: 257 / 108

2022

Week 37, Month 9

18 صفر ١٤٤٤/٢

8 items كود بعد عدد السلسلة في قائمة

acc-num و اسند ذلك الرقم الى متغير

تالي 1: استخدم عطف لراكم

تالي 2: اياك in تنجس ما ذالك في سلسلة لفرقة مصورة

صن السلسلة

10

items = ["whirring", "wow!", "calendar", "wry", "nglass",

"", ""]

acc-num = 0

for item in items:

if "w" in item:

1 acc-num += 1

Print (acc-num)

2

⇒ (4)

3

4 Sentence وبتاي كوي ابا "a" و "e"

num-a-or-e وخرن الباع في كوي

5 طريقة 1: ادمع كل كلمة في السلسلة وبتاي كوي كل من a و e

طريقة 2: استخدم in

طريقة 3: عليك استخدام or و elif

6 sentence = "Python is a high level general purpose

programming ----"

7 sentence-words = sentence.split()

num-a-or-e = 0

8 for word in sentence-words:

if ("a" in word) or ("e" in word):

num-a-or-e += 1

Print (num-a-or-e)

⇒ (11)

Tuesday 13
Mardi

September - Septembre

الثلاثاء ١٣

أيلول - سبتمبر

17 Safar 2 / 1444

Day: 256 / 109

Week 37, Month 9

17 صفر ١٤٤٤ / ٢

* فيا لي كيات ططورا في مشيت ما / بطول ايام عدد ططورا بكرة
8 `num-rainy-months` من 3.0 مل وكمتر غيا في الشهر
`rainfall_mi = "1.65, 1.46, 2.05, 3.03, 3.35, 3.46, 2.83,`
9 `3.23, 3.5, 2.52, 2.8, 1.85"`
`rainfall_mi_list = rainfall_mi.split(",")`
10 `num-rainy-months = 0`
For x in `rainfall_mi_list`:
11 if `float(x) > 3.0`:
`num-rainy-months += 1`
12 `print(num-rainy-months)`
1 $\Rightarrow (5)$

* آلفير Sentence كوي ل String ، آلفير كود كرد عدد ططورا
Sentence تبا و تبا نفس الحرف و تبا نفس الحرف و تبا نفس الحرف
3 `same-letter-count` و تبا نفس الحرف
Sentence = "students Flock to the arb for a variety
4 of outdoor..."
`sentence_words = sentence.split()`
5 `same-letter-count = 0`
For word in `sentence_words`:
6 if `(word[0] == word[-1]) or (len(word) == 1)`:
`same-letter-count += 1`
7
8 `print(same-letter-count)`
 $\Rightarrow (2)$

9
SEP

* what is printed by the following statements?

8

s = "we are learning!"

9

x = 0

for i in s:

10

if i in ['a', 'b', 'c', 'd', 'e']:

x += 1

11

print(x)

12

⇒ 5

1

+ what is printed by the following statements?

2

list = [5, 2, 1, 4, 9, 10]

min_value = 0

3

for item in list:

if item < min_value:

4

min_value = item

print(min_value)

5

⇒ 0

6

min_value was set to a number that was smaller than any of the numbers in the list, so it was never updated in the for loop.

7

8

min_value = list[0]

min_value = 0

Sunday 11
Dimanche

September - Septembre

الأحد ١١

أيلول - سبتمبر

15 Safar 2/1444

Day: 254 / 111

2022

Week 37, Month 9

١٥ صفر ١٤٤٤/٢

s = "what if we went to the Zoo"

8 x = 0

for i in s:

9 if i in ['a', 'e', 'i', 'o', 'u']:

x += 1

10 print(x)

11 ⇒ 8

12 word = "onomatopoeia"

Goal: count the number of o's in word

1

o_count = 0

2

for c in word:

3

if c == 'o':

o_count = o_count + 1

4

print("There are " + str(o_count) + " o's in " + word)

5

6

⇒ There are 4 o's in onomatopoeia

7

nums = [9, 3, 8, 11, 5, 29, 2]

best_num = nums[0]

8

for n in nums:

if n > best_num:

best_num = n
print(best_num)

⇒ 29

9
SEP

8 str1 = "Today you are you! That is truer than true?
There is no one alive who is you-er than you!"

9 if "false" in str1:
output = "False, you aren't you?"
10 elif "true" in str1:
output = "True! You are you!"
11 else:
output = "Neither true nor false!"

12 The Accumulator pattern with Conditionals &
1 Accumulating a maximum value

2 * The Accumulator pattern with conditionals

3 phrase = "what a wonderful day to program"
tot = 0 # initialise acc var
for char in phrase: # iterate through seq
4 if char != " ":
tot = tot + 1 # update acc var
5 print(tot) # acc var has answer

6 ⇒ 26

7 Ⓢ ← phrase = "أهلاً بك!"
3 ← phrase = "XYZ "

8

Chained conditionals

8

| | | | |
|------------|------------|------------|------------|
| code block | code block | code block | code block |
| if | if | if | if |
| code block | else | elif | elif |
| | code block | else | elif |
| | | code block | code block |

11

* what will be the following code print if
 $x=3, y=5$, and $z=2$?

12

```

1   if 3x < 5y and 3x < 2z:
2       print("a")
3   elif 5y < 3x and 2y < 2z:
4       print("b")
5   else:
6       print("c")

```

} ⇒ C

4

* create one conditional to find whether "false" is in string str1, if so, assign variable output the string "False! you aren't you?" check to see if "true" is in string str1 and if it is then assign "True! you are you!" to the variable output. if neither are in str1, assign "neither true nor false" to output.

8

* will the following code cause an error? $\Rightarrow$ No

```

8
x = -10
9
if x < 0:
    print("The negative number", x, "is not valid here.")
10
else:
    if x > 0:
11
        print(x, "is a positive number")
    else:
12
        print(x, "is 0")

```

* Chained conditionals

```

1
x = 10
2
y = 10
3
if x < y:
    print("x is less than y")
4
else:
    if x > y:
5
        print("x is greater than y")
    else:
6
        print("x and y must be equal")

```

```

7
x = 10
y = 10
8
if x < y:
    print("x is less than y")
elif x > y:
    print("x is greater than y")
else:
    print("x and y must be equal")

```

$\Rightarrow$ x and y must be equal

* Will the following code cause an error? $\Rightarrow$ Yes

8

↓ لأنه لو لم أكتب else

$x = -10$

9

if $x < 0$:

print("The negative number, x, is not valid")

10

else:

print(x, "is a positive number")

if x (else)

print("This is always printed")

else
أليس كذلك؟
في كل مرة

1

* Nested Conditionals

2

$x = 10$

$y = 10$

3

if $x < y$:

10 < 10

4

print("X is less than y")

else:

5

if $x > y$:

print("X is greater than y")

6

else:

print("X and Y must be equal")

7

8

$\Rightarrow$ X and Y must be equal

~~X is less than Y $\Leftarrow$ $x \geq 10$ $x \geq 5$ $x \geq 0$~~
~~X is greater than Y $\Leftarrow$ $y \geq 3$ $x \geq 10$~~

Tuesday
Mardi

6

الثلثاء ٦

September - Septembre

أيلول - سبتمبر

10 Safar 2/1444

Day: 249 / 116

2022

Week 36, Month 9

10 صفر ١٤٤٤/٢

* write code to assign the string "You can apply to SI!"
to output if the string "SI 106" is in the list
courses. if it is not in courses, assign the value
"Take SI 106!" to the variable output.

Courses = ["ENGR 101", "SI 110", "ENG 125", "SI 106",
"CHEM 130"]

if "SI 106" in Courses:
output = "You can apply to SI!"
else:
output = "Take SI 106!"

Unary selection, nested conditionals, & chained
conditionals

* omitting the clause: Unary selection

x = 10
if x < 0:
print("The negative number", x, "is not valid here")
print("This is always printed")

⇒ This is always printed

← x = -10 ~ 6/5/21
The negative number -10 is not valid here
This is always printed

Conditional Execution

8

$X=15$

9

if $X \% 2 == 0$:

print(X, "is even")

10

else:

print(X, "is odd")

11

print("IT IS ODD")

} $\Rightarrow$ 15 is odd
IT IS ODD

How many lines of code can appear in indented code block below the if and else lines in a conditional?

12

1 - one or more

2

if $(4+5 == 10)$:

print("TRUE")

3

else:

print("FALSE")

} $\Rightarrow$ FALSE

4

if $(4+5 == 10)$:

print("TRUE")

5

else:

print("FALSE")

6

print("TRUE")

} $\Rightarrow$ FALSE
TRUE

7

8

The in & not in operators

8

print('p' in 'apple') $\Rightarrow$ True

9

print('i' in 'apple') $\Rightarrow$ False

print('ap' in 'apple') $\Rightarrow$ True

10

print('pa' in 'apple') $\Rightarrow$ False

11

print('a' in 'a') $\Rightarrow$ True

print('apple' in 'apple') $\Rightarrow$ True

12

print(' ' in 'a') $\Rightarrow$ True

print(' ' in 'apple') $\Rightarrow$ True

1

print('X' not in 'apple') $\Rightarrow$ True

2

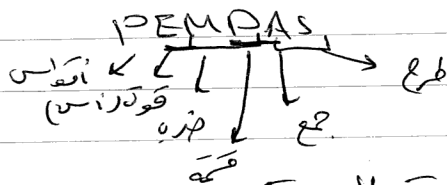
print('a' in ["apple", "absolutely", "application", "unhappy"]) $\Rightarrow$ False

3

Precedence of operators

4

5



6

مع انقاس في اولى العمليات

7

$3 \times 5 > 10$ and $4 + 6 \geq 11$

$((3 \times 5) > 10 \text{ and } ((4 + 6) \geq 11))$

8

$15 > 10$ and $4 + 6 \geq 11$

True and $4 + 6 \geq 11$

True and $10 \geq 11$

True and False

False

النتيجة

Boolean values and Boolean Expressions

Boolean Expressions: ~~العبارة~~ ^{العبارة}

8

* True

* $3 == 4$

9

* $3 + 4 == 7$

10

Logical operators

11

Literal

Comparison

Logical operators

12

True

False

$L == R$

$L != R$

$L < R$

$L <= R$

$L > R$

$L >= R$

L and R

L or R

not B

1

2

| and | L True | L False |
|---------|--------|---------|
| R True | True | False |
| R False | False | False |

| OR | L True | L False |
|---------|--------|---------|
| R True | True | True |
| R False | True | False |

4

| | B True | B False |
|-----|--------|---------|
| not | False | True |

5

جواب

$X = 5$ True True

6

print($X > 0$ and $X < 10$) $\Rightarrow$ True

7

$n = 25$

Print($n \% 2 == 0$ or $n \% 3 == 0$) $\Rightarrow$ False

8

False False

Boolean Expressions

8

Print(False) $\Rightarrow$ False

9

Print(True) $\Rightarrow$ True

Print(type(True)) $\Rightarrow$ <class 'bool'>

10

Print(type("True")) $\Rightarrow$ <class 'str'>

Print(5 == 5) $\Rightarrow$ True

11

Print(5 == 6) $\Rightarrow$ False

* True و False

12

$(L) == (R)$ True if (L) equal to (R)
 $(L) != (R)$ if (L) not equal to (R)

1

2

Print(5 != 6) $\Rightarrow$ True

3

$(L) == (R)$

$(L) != (R)$

4

$(L) < (R)$

$(L) <= (R)$

5

$(L) > (R)$

$(L) >= (R)$

6

Print(3 < 4) $\Rightarrow$ True

7

Print(4 <= 4) $\Rightarrow$ True

Print(5 > 6) $\Rightarrow$ False

8

Print(7 >= 8) $\Rightarrow$ False

* what is the type of your iterable?

8 t = ("coach", "chair", "washer", "dryer", "table")
9 For z in t:
print(z) } ⇒ tuple

10 * what is the iterator variable?

11 t = ["coach", "chair", "washer", "dryer", "table"]
12 For z in t:
print(z) } ⇒ String
Every item in the iterator variable is string

1 * what's the type of iterator variable in the first iteration?

2 t = [9, "setter", 3, "wing spiker", 18, "middle blocker"]
3 For z in t:
print(z) } ⇒ integer

9 SEP

4 16 - 2 * 5 // 3 + 1
5 10
3
13
14
6 ⇒ 14

7 X = 15
Y = X
8 X = 22) ⇒ X is 22 and Y is 15